

Solar Aware Multi-Sink Data Aggregation Protocol for Wireless Sensor Assisted IOT

Dissertation on the B. Tech final year project is being submitted towards partial fulfillment of the requirement for the Degree of

Bachelor of Technology
in
Information Technology

from Maulana Abul Kalam Azad University of Technology, West Bengal

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June – 2026

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FORWARD

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ACKNOWLEDGEMENT

We express our sincere gratitude to **Mrs. Moumita Deb** of Department of Information Technology, RCC Institute of Information Technology, Kolkata for extending her valuable times for us to take up this problem as our final year project.

We are also indebted to **Dr. Soumyadip Dhar(HOD)** , **Dr Hiranmoy Roy** for their unconditional help and inspiration.

Last but not the least we would like to express our gratitude to **Mrs. Moumita Deb** of our department who helped us in their own way whenever needed.

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Certificate of Acceptance

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Name of the Examiner

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Table of Contents

Abstract

CHAPTER	Page No
CHAPTER - 1	
1.1 INTRODUCTION	2
1.2 LITERATURE REVIEW	3
CHAPTER-2	
2.1 WORK FLOW	8
2.2 METHODOLOGIES OF IMPLEMENTATION	12
2.3 SOFTWARE & HARDWARE REQUIREMENTS	16
2.4 PROPOSED WORK	20
CHAPTER-3	
3.1 TEST CASES & SYSTEM VALIDATION	24
3.2 OBSERVED OUTPUT	30
3.3 PERFORMANCE ANALYSIS	42
CHAPTER-4	
4.1 CONCLUSION	51
4.2 CHALLENGES	51
4.3 FUTURE SCOPE	52
REFERENCES	54
APPENDIX (Program Code)	58

ABSTRACT

Wireless Sensor Networks (WSNs) integrated with Internet of Things (IoT) systems are widely used in applications such as smart agriculture, smart buildings, environmental monitoring, and industrial automation. However, sensor nodes generally operate with limited battery power, causing rapid energy depletion and reducing network lifetime. Traditional routing and cluster-based protocols often fail to consider dynamic energy conditions, solar energy availability, and efficient load balancing across the network. To address these challenges, this project proposes a Solar-Aware Multi-Sink Data Aggregation Protocol for Wireless Sensor-Assisted IoT. The proposed protocol combines solar energy harvesting, data aggregation, multi-sink communication, and Genetic Algorithm (GA)-based optimization to improve overall network performance and energy efficiency. The protocol dynamically selects Cluster Heads (CHs) and Multi-Sink Cluster Heads (MS-CHs) based on multiple parameters including residual energy, solar harvesting state, geographical position, and distance to the Base Station (BS). Sensor nodes transmit data to CHs, where data aggregation is performed, and selected MS-CH nodes forward aggregated packets to the base station through optimized routes. The Genetic Algorithm continuously evaluates possible routing and sink-selection solutions to adapt to changing energy conditions and network states. Performance evaluation metrics such as network lifetime, packet delivery ratio (PDR), residual energy, convergence speed, latency, and throughput are used for analysis. The proposed approach aims to minimize energy consumption, balance network load, and significantly increase network lifetime compared with conventional protocols such as LEACH and shortest-path-based methods.

CHAPTER- 1

Introduction:-

The rapid development of the Internet of Things (IoT) has increased the demand for efficient and intelligent communication among connected devices. Wireless Sensor Networks (WSNs) play a vital role in IoT systems by collecting and transmitting environmental information such as temperature, humidity, pressure, motion, and other sensor data. These sensor nodes are commonly deployed in large-scale applications including smart cities, precision agriculture, industrial monitoring, healthcare systems, and environmental observation systems. Despite their importance, sensor nodes are constrained by limited battery capacity, making energy efficiency one of the most critical challenges in WSN design.

Traditional routing protocols in WSNs generally focus on shortest-path communication or random cluster-head selection mechanisms. Approaches such as LEACH use random cluster-head election without considering node energy conditions, while shortest-path methods ignore important factors such as energy balance and solar energy availability. As a result, certain nodes become overloaded and exhaust their energy rapidly, creating network holes and reducing overall network lifetime.

Energy harvesting techniques, especially solar energy harvesting, provide a promising solution to this problem by enabling sensor nodes to replenish energy from environmental sources. However, simply adding solar-powered nodes is insufficient because routing decisions must intelligently adapt to the dynamic energy state of each node. Therefore, routing protocols should consider both current residual energy and potential future energy availability through solar harvesting.

Data aggregation is another important technique used in WSNs to reduce unnecessary communication overhead. Instead of transmitting all raw sensor data directly to the base station, sensor nodes send information to Cluster Heads (CHs), which aggregate and compress data before forwarding it. This reduces redundant transmissions and significantly saves energy.

In addition, the use of multiple sink nodes can further improve communication efficiency and reduce transmission distance. In the proposed work, a Multi-Sink Cluster Head (MS-CH) mechanism is introduced where an optimized relay node is selected to handle long-distance communication to the Base Station. The selection of CHs and MS-CHs is performed using a Genetic Algorithm (GA) that evaluates multiple routing possibilities and chooses the best solution based on several parameters including residual energy, solar harvesting rate, node position, load balancing, and distance to the Base Station.

The proposed Solar-Aware Multi-Sink Data Aggregation Protocol for Wireless Sensor-Assisted IoT aims to create an adaptive and intelligent routing framework capable of responding to changing network conditions. By integrating solar awareness, multi-sink routing, data aggregation, and evolutionary optimization, the protocol seeks to improve energy efficiency, increase packet delivery performance, reduce communication delay, and maximize overall network lifetime.

1.2 Literature Review

Wireless Sensor Networks (WSNs) have attracted significant attention in recent years due to their wide range of applications in Internet of Things (IoT) environments, including environmental monitoring, smart agriculture, healthcare systems, industrial automation, smart buildings, and smart city infrastructures. In these systems, sensor nodes collect and transmit real-time information such as temperature, humidity, pressure, and motion data. However, traditional WSNs are mainly composed of battery-powered sensor nodes with limited energy resources, creating challenges such as rapid energy depletion, reduced network lifetime, communication failures, and poor network reliability. Initial research in this field primarily focused on reducing communication overhead and developing energy-efficient routing methods. Although these approaches improved network performance to some extent, they often failed to provide long-term sustainability because sensor nodes exhausted their battery energy quickly.

Solar Energy Harvesting in WSNs

To address the limitations of battery-powered sensor networks, researchers introduced energy harvesting techniques that enable nodes to collect renewable energy from environmental sources. Among these techniques, solar energy harvesting has emerged as one of the most practical and widely used solutions due to its availability and sustainability. Solar-powered WSNs allow sensor nodes to recharge using sunlight, thereby extending network lifetime and reducing dependence on battery replacement. Existing studies on Energy Harvesting Wireless Sensor Networks (EH-WSNs) indicate that adaptive energy management strategies combined with solar power can significantly improve network performance and reduce node failures. However, many existing solar-aware protocols still rely on static routing and fixed clustering mechanisms, which cannot effectively adapt to continuously changing solar energy availability and varying node energy conditions.

Data Aggregation and Clustering Techniques

Data aggregation has become an essential technique for reducing communication overhead and improving energy efficiency in WSNs. Instead of transmitting all sensed data directly to the Base Station (BS), sensor nodes send information to Cluster Heads (CHs), which aggregate and compress the collected data before forwarding it. This process significantly reduces redundant transmissions and conserves energy. To support efficient data aggregation, clustering techniques have been widely adopted. Protocols such as LEACH introduced randomized cluster-head rotation to distribute energy consumption among nodes. While LEACH successfully reduced communication cost, random CH selection frequently resulted in unstable clusters and inefficient energy utilization.

To improve cluster-head selection, researchers later incorporated intelligent optimization methods such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Simulated Annealing, and fuzzy logic approaches. Among these optimization methods, GA-based clustering demonstrated strong performance due to its capability of dynamically

selecting optimal cluster heads using multiple parameters such as residual energy, node distance, communication load, and network topology characteristics.

Multi-Sink Architectures in WSNs

Traditional WSNs generally employ a single sink or base station to collect information from all sensor nodes. However, single-sink architectures often suffer from hotspot problems where nodes located close to the sink experience excessive communication load and deplete their energy faster. To overcome these limitations, researchers proposed multi-sink architectures that distribute traffic among multiple collection points or relay nodes. Multi-sink communication reduces transmission distance, balances communication load, decreases congestion, and improves network reliability.

Studies have demonstrated that multi-sink routing significantly improves network performance by increasing throughput, reducing packet loss, minimizing latency, and extending network lifetime. However, many existing multi-sink approaches focus only on routing efficiency and fail to consider dynamic energy conditions and solar energy availability.

Genetic Algorithm-Based Optimization in WSNs

Genetic Algorithms have gained significant popularity for solving complex optimization problems in WSNs because of their ability to simultaneously optimize multiple objectives. GA-based approaches operate by generating a population of possible solutions, evaluating them using a fitness function, and continuously improving solutions through selection, crossover, and mutation operations. Research findings indicate that GA-based routing and clustering methods effectively identify near-optimal communication paths, minimize redundant transmissions, balance energy consumption, and adapt dynamically to changing network conditions.

Several studies utilize fitness functions based on factors such as residual energy, communication distance, delay, load balancing, and packet delivery performance. In solar-aware systems, additional parameters such as solar harvesting state and future energy availability can further improve routing decisions. However, previous research rarely integrates GA optimization with solar energy awareness, dynamic cluster-head selection, data aggregation, and multi-sink communication within a unified framework.

Comparative Analysis of Existing Approaches

Shortest-path algorithms such as Dijkstra primarily focus on minimizing path distance and often ignore energy conditions, solar state, and network load balancing. LEACH performs random cluster-head selection without considering node energy conditions, resulting in inefficient resource utilization. Greedy algorithms generally optimize local decisions and may fail to achieve globally optimal routing solutions. Simulated Annealing and Particle Swarm Optimization methods provide improved optimization performance but may suffer from slower convergence or increased computational complexity in large-scale WSN environments.

Compared to these techniques, Genetic Algorithms provide advantages such as handling large search spaces, supporting multiple optimization objectives, adapting to changing network states, and selecting energy-efficient routes dynamically.

Research Gap

Although significant advancements have been achieved in solar-powered WSNs, intelligent clustering techniques, data aggregation mechanisms, Genetic Algorithm-based optimization, and multi-sink architectures, most studies have investigated these technologies independently. Very limited research integrates **solar energy harvesting, dynamic cluster-head selection, multi-sink communication, data aggregation, and GA-based optimization** within a single adaptive framework.

Therefore, there exists a strong need for an integrated system capable of intelligently selecting Cluster Heads (CHs) and Multi-Sink Cluster Heads (MS-CHs) based on residual energy, solar harvesting state, geographical position, and communication distance. Such a framework can improve sustainability, enhance scalability, reduce energy consumption, balance network load, and maximize network lifetime for Wireless Sensor-Assisted IoT environments.

Table 1.1 Comparative Studies

Sl. No.	Author(s)	Paper Title	Main Contribution	Challenges Identified / Addressed
1	Yin Wu, Wenbo Liu	<i>Routing Protocol Based on Genetic Algorithm for Energy Harvesting WSN (EHGUC–OAPR)</i>	- Proposed EHGUC , a GA-based unequal clustering mechanism with OAPR adaptive routing . - Designed a multi-objective fitness function considering distance, energy sustainability, and intra-cluster communication cost. - Used centralized clustering for energy-harvesting sensor networks.	- Traditional routing protocols perform poorly in energy-harvesting environments. - Uneven solar harvesting causes imbalance in CH energy consumption. - Requirement for maintaining energy sustainability rather than only maximizing network lifetime. - Need for reliable multihop routing while preventing node shutdown.
2	Zeli Xue	<i>Routing Optimization of Sensor Nodes in IoT Based on Genetic Algorithm</i>	- Proposed a multi-objective GA for node selection and routing optimization. - Introduced similarity-based crossover and new-individual injection to avoid local optima. - Achieved high network coverage while minimizing active nodes and balancing	- Node redundancy and coverage holes affect IoT networks. - Repeated node usage causes premature energy depletion. - Nodes near sinks suffer overload problems. - Difficulty in optimizing multiple parameters simultaneously.

Sl. No.	Author(s)	Paper Title	Main Contribution	Challenges Identified / Addressed
			energy consumption.	
3	Padmalaya Nayak, Bhavani Vathasavai	<i>Genetic Algorithm Based Clustering Approach for WSN to Optimize Routing (CONFLUENCE 2017)</i>	- Used Genetic Algorithm for cluster-head selection based on residual energy and distance. - Improved LEACH by eliminating random CH selection. - Increased network lifetime and energy stability.	- Random CH selection in LEACH creates unstable clusters. - Uneven energy dissipation among nodes. - CH placement near boundaries increases communication cost.
4	A. B. Alnajjar et al.	<i>Wireless Sensor Network Optimization Using Genetic Algorithm (JRC 2022)</i>	- Proposed a GA using 2D integer encoding and special crossover operations for optimized CH placement. - Considered operating energy, communication energy, and residual energy. - Achieved optimal cluster distribution and coverage.	- Excessive or insufficient CH numbers affect network performance. - Poor CH placement may create isolated nodes. - Need to maintain communication limits and avoid CH overload.
5	Ataul Bari et al.	<i>A Genetic Algorithm Based Approach for Energy Efficient Routing in Two-Tiered Sensor Networks</i>	- Applied GA for optimal relay node routing to improve network lifetime. - Introduced mutation techniques targeting critical high-energy nodes. - Achieved better performance than traditional approaches.	- Integer Linear Programming (ILP) becomes impractical for large WSNs. - Relay node failures create network-wide effects. - Need routing based on transmission and reception energy.
6	Innocent Njini, O. Ekabua	<i>Genetic Algorithm Based Energy Efficient Optimization Strategies in WSN: A Survey</i>	- Surveyed GA applications in clustering, routing, and network coverage optimization. - Identified strengths of GA in handling complex optimization problems and avoiding local optima.	- Communication overhead increases energy consumption. - Multi-hop communication creates bottlenecks near sink nodes. - Need accurate prediction models and reduced overhead.
7	Hosam El-Ocla et al.	<i>Routing Using Genetic Algorithm in Wireless Sensor Network (2020)</i>	- Applied GA for generating multiple routing paths. - Compared GA with Dijkstra, AODV, and other routing methods. - Improved throughput, fault tolerance,	- Traditional shortest-path algorithms fail under node failures. - AODV causes additional energy consumption due to route discovery. - Mobility creates

Sl. No.	Author(s)	Paper Title	Main Contribution	Challenges Identified / Addressed
			and energy efficiency.	packet loss and route failures.
8	Shalli Rani et al.	<i>Dynamic Clustering Approach Based on WSN Genetic Algorithm for IoT Applications (Wireless Networks 2020)</i>	- Proposed dynamic GA-based clustering for IoT environments. - Fitness function included distance, energy cost, and cluster parameters. - Introduced relay nodes for better performance.	- Large-scale IoT environments require scalable and reliable clustering. - Existing approaches struggle with dynamic loads and heterogeneity. - Need to avoid cluster instability and CH overload.
9	Proposed Work	Solar-Aware Multi-Sink Data Aggregation Protocol for Wireless Sensor Assisted IoT	- Integrates solar energy harvesting, dynamic data aggregation, multi-sink routing, and Genetic Algorithm optimization in a unified framework. - Selects CH and MS-CH using residual energy, solar state, geographical position, and distance to BS. - Uses data aggregation through CH → MS-CH → BS architecture to reduce communication overhead. - Dynamically re-elects nodes according to changing network conditions.	- Existing studies largely treat solar awareness, multi-sink routing, clustering, and optimization independently. - Need adaptive routing for changing energy conditions. - Requirement for load balancing and energy sustainability. - Need to maximize network lifetime while improving PDR, throughput, and reducing delay.

CHAPTER- 2

2.1. Workflow

The proposed **Solar-Aware Multi-Sink Data Aggregation Protocol for Wireless Sensor-Assisted IoT** is designed to improve network lifetime, energy efficiency, load balancing, and communication reliability in Wireless Sensor Networks (WSNs). Traditional WSN protocols such as LEACH utilize clustering techniques to reduce communication cost; however, they suffer from several limitations including random Cluster Head (CH) selection, lack of energy awareness, absence of solar-energy consideration, and direct communication between Cluster Heads and the Base Station (BS). These limitations result in rapid energy depletion, communication bottlenecks, packet loss, and reduced network lifetime.

To overcome these limitations, the proposed protocol integrates **solar energy harvesting, cluster- head based communication, dynamic Multi-Sink architecture, data aggregation, and Genetic Algorithm (GA)-based optimization** into a unified framework. The protocol intelligently selects Cluster Heads and Multi-Sink Cluster Heads according to network conditions and energy availability.

Initially, sensor nodes are randomly deployed in the monitoring area to collect environmental information such as temperature, humidity, motion, and other sensing parameters. Since sensor nodes have limited battery resources, each node is equipped with a **solar energy harvesting mechanism** that enables battery recharging using available solar energy. Unlike conventional approaches that consider only current battery status, the proposed protocol considers both **residual energy and future energy availability through solar harvesting**.

After node deployment and energy initialization, the network forms communication clusters. Rather than selecting Cluster Heads randomly as performed in LEACH, the proposed protocol applies a **Genetic Algorithm** to select optimal Cluster Heads based on multiple parameters, including:

- Residual energy of sensor nodes
- Solar energy harvesting status
- Node geographical position
- Communication distance
- Load balancing conditions

Following Cluster Head selection, ordinary sensor nodes join the nearest CH to establish communication clusters.

During network operation, some Cluster Heads may become overloaded or experience rapid energy depletion due to excessive communication demands. Such situations create bottlenecks, increase packet loss, and reduce communication efficiency. To address this issue, the proposed system dynamically identifies problematic Cluster Heads and applies **Genetic Algorithm optimization to select a Multi-Sink Cluster Head (MS-CH)**.

The MS-CH acts as an intermediate relay node between Cluster Heads and the Base Station. Unlike traditional approaches where Cluster Heads directly communicate with the BS, the proposed communication path follows:

**Sensor Node → Cluster Head (CH) → Multi – Sink Cluster Head (MS – CH) (if need)
→ Base Station (BS)**

The operational process is performed as follows:

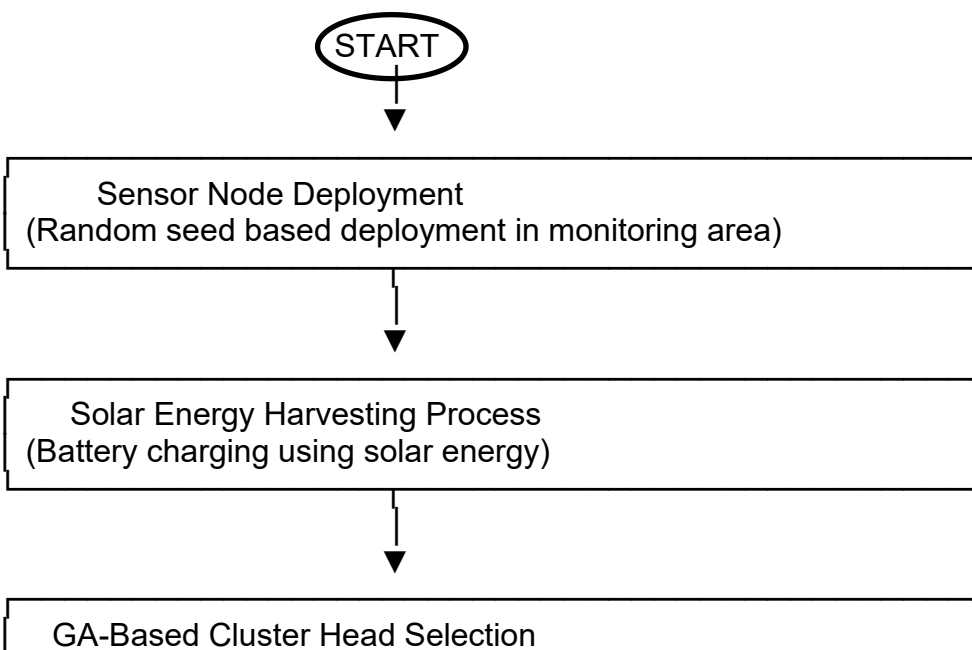
- Sensor nodes transmit sensed information to their assigned Cluster Head.
- Cluster Heads aggregate and compress received packets to eliminate redundant information.
- The selected MS-CH receives aggregated data from multiple Cluster Heads.
- MS-CH forwards optimized packets to the Base Station.

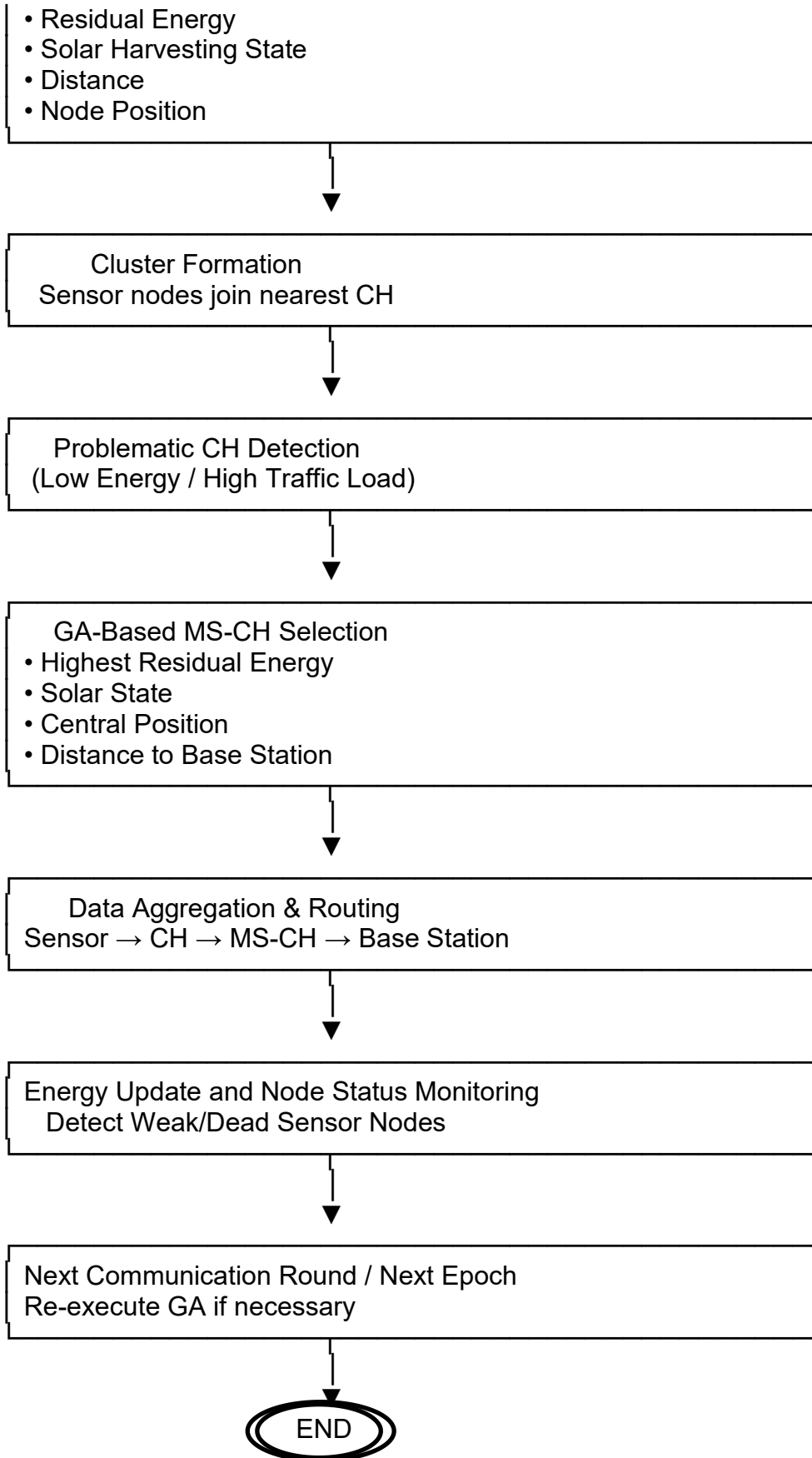
By reducing long-distance communication and balancing network traffic, the protocol significantly reduces energy consumption and improves network stability.

After completion of each communication round, the residual energy and operational status of all sensor nodes are updated. If the energy level of the selected MS-CH decreases below a predefined threshold or significant network changes occur, the Genetic Algorithm automatically re-executes to select new Cluster Heads and Multi-Sink Cluster Heads.

Through the integration of **solar-awareness, data aggregation, dynamic Multi-Sink routing, and Genetic Algorithm optimization**, the proposed workflow effectively minimizes energy consumption, improves packet delivery ratio, reduces communication delay, balances network load, and substantially increases overall network lifetime in Wireless Sensor-Assisted IoT environments.

Fig. 3.1 Workflow Diagram





Workflow Steps

Step 1: Sensor Node Deployment

Sensor nodes are randomly deployed within the target monitoring area to collect environmental information and establish the initial network topology.

Step 2: Solar Energy Harvesting

Each sensor node harvests solar energy from the environment to recharge its battery and improve long-term network sustainability.

Step 3: Genetic Algorithm-Based Cluster Head Selection

The Genetic Algorithm dynamically selects optimal Cluster Heads according to:

- Residual energy
- Solar energy harvesting status
- Node location
- Communication distance
- Load balancing requirements

Step 4: Cluster Formation

Sensor nodes associate with their nearest Cluster Head and establish communication clusters.

Step 5: Problematic Cluster Head Detection

The system continuously monitors Cluster Head performance and identifies overloaded or low-energy CHs.

Step 6: Genetic Algorithm-Based Multi-Sink Selection

The Genetic Algorithm selects an optimal Multi-Sink Cluster Head based on:

- Residual energy
- Solar harvesting status
- Geographical position
- Distance to Base Station

Step 7: Data Aggregation and Routing

The communication process follows:

Sensor → CH → MS – CH (if needed) → BS

where:

- Sensor nodes transmit data to CHs.
- CHs aggregate and compress collected data.
- MS-CH forwards optimized data packets.
- BS receives final information.

Step 8: Energy Update and Node Status Monitoring

After each communication cycle, node energy levels and operational conditions are updated.

Step 9: Next Epoch and Dynamic Re-election

If network conditions change or the MS-CH energy level falls below the threshold value:

- The Genetic Algorithm re-executes.
- New CH and MS-CH nodes are selected.
- Routing paths are updated dynamically.

This process continues iteratively to maintain energy-efficient and reliable communication throughout the network lifetime.

2.2 Methodologies of Implementation

The proposed **Solar-Aware Multi-Sink Data Aggregation Protocol for Wireless Sensor-Assisted IoT** is implemented using an integrated methodology that combines **solar energy harvesting, cluster-based communication, dynamic Multi-Sink architecture, data aggregation, and Genetic Algorithm (GA)-based optimization**. The implementation methodology aims to improve network lifetime, reduce energy consumption, balance communication load, and enhance the reliability of data transmission within Wireless Sensor Networks (WSNs). The implementation process is divided into several stages as described below.

Network Model

The proposed system considers a Wireless Sensor Network (WSN) composed of **N sensor nodes** randomly deployed within a two-dimensional square sensing field. All sensor nodes are assumed to be static after deployment and initialized with identical energy levels to maintain uniform simulation conditions.

The network architecture consists of:

- Sensor Nodes (SN)
- Cluster Heads (CH)
- Multi-Sink Cluster Heads (MS-CH)
- Base Station (BS)

The Base Station is placed at a predefined location within or near the network area for centralized monitoring and data collection. Each sensor node operates within a fixed communication range

and continuously senses environmental parameters such as temperature, humidity, pressure, and motion information.

Unlike conventional WSNs where each node directly transmits information to the Base Station, the proposed protocol adopts a hierarchical communication structure:

Sensor Node → Cluster Head (CH) → Multi-Sink Cluster Head (MS-CH) → Base Station (BS)

This communication architecture reduces transmission distance and improves energy utilization.

Energy Model

The proposed protocol adopts the **First-Order Radio Energy Model** to estimate energy consumption during communication operations. The total energy consumed by sensor nodes consists of:

- Transmission Energy ((E_{tx}))
- Reception Energy ((E_{rx}))
- Data Aggregation Energy ((E_{DA}))

The energy consumption for transmission depends on communication distance and packet size.

The protocol incorporates **solar energy harvesting** to improve network sustainability. Each sensor node continuously replenishes energy using available solar energy. The harvested energy dynamically changes according to environmental conditions such as daylight intensity and solar availability.

The residual energy of a sensor node after each communication cycle can be represented as:

$$[E_{residual} = E_{initial} + E_{harvested} - (E_{tx} + E_{rx} + E_{DA})]$$

Where:

- $(E_{initial})$ = Initial battery energy
- $(E_{harvested})$ = Harvested solar energy
- (E_{tx}) = Transmission energy
- (E_{rx}) = Reception energy
- (E_{DA}) = Data aggregation energy

To simulate real-world conditions, controlled variations and environmental uncertainties are introduced into the energy harvesting process.

Cluster Head Selection

To ensure balanced energy utilization and efficient communication, Cluster Heads are selected dynamically rather than randomly as performed in traditional protocols such as LEACH.

The proposed protocol utilizes a **Genetic Algorithm-based Cluster Head Selection mechanism** where multiple performance parameters are evaluated, including:

- Residual energy of the node
- Solar harvesting status
- Node geographical position
- Communication distance
- Local node density
- Load balancing criteria

Only nodes whose residual energy exceeds a predefined threshold value are considered eligible for becoming Cluster Heads.

The Genetic Algorithm generates multiple candidate solutions and evaluates them using a fitness function. The optimal Cluster Head set is selected based on the highest fitness value.

Cluster Formation and Data Aggregation

Following Cluster Head selection, ordinary sensor nodes associate with their nearest Cluster Head to establish communication clusters.

The communication process occurs in the following stages:

1. Sensor nodes collect environmental information.
2. Sensor nodes transmit data to their assigned Cluster Head.
3. Cluster Heads receive and aggregate data packets.
4. Redundant information is removed through data compression techniques.

Data aggregation reduces:

- Communication overhead
- Redundant transmissions
- Energy consumption

This process significantly improves network efficiency.

Problematic Cluster Head Identification

During network operation, some Cluster Heads may become overloaded or experience unfavorable communication conditions that negatively affect network performance.

A Cluster Head is identified as problematic when one or more of the following conditions occur:

- Residual energy falls below a threshold level

- The Cluster Head is located far from the Base Station
- Transmission energy requirements become excessive
- Estimated packet loss probability becomes high
- Communication delay increases
- The Cluster Head manages a large number of member nodes

Identifying problematic Cluster Heads enables adaptive routing decisions and prevents network bottlenecks.

Multi-Sink Selection Using Genetic Algorithm

The proposed protocol introduces a **dynamic Multi-Sink mechanism** to improve communication efficiency and reduce the burden on overloaded Cluster Heads.

Instead of allowing all Cluster Heads to directly communicate with the Base Station, a **Multi-Sink Cluster Head (MS-CH)** is selected using a Genetic Algorithm.

The Genetic Algorithm evaluates candidate nodes according to:

- Residual energy
- Solar energy harvesting status
- Distance to Base Station
- Node centrality
- Communication load
- Hop count
- Packet loss probability
- The fitness function attempts to minimize:
- Total energy consumption
- Packet loss
- Communication delay
- Hop count
- Load imbalance

The Genetic Algorithm performs the following operations:

- Population initialization
- Fitness evaluation
- Selection
- Crossover
- Mutation
- Optimal solution generation

The selected MS-CH acts as an intermediate relay node.

The optimized communication path becomes:

Sensor Node → CH → MS – CH(if needed) → BS

This Multi-Sink mechanism:

- Reduces long-distance transmission cost
- Balances network traffic
- Minimizes packet loss
- Improves throughput
- Extends network lifetime

Energy Update and Dynamic Re-election

After each communication round, the protocol updates the status of all sensor nodes.

The system performs:

- Residual energy calculation
- Dead node identification
- Solar energy updates
- Network performance monitoring

If:

- CH energy drops below threshold
- MS-CH energy decreases significantly
- Communication performance degrades

the Genetic Algorithm automatically re-executes to dynamically select new CH and MS-CH nodes.

This adaptive mechanism ensures continuous network operation and maintains stable communication performance throughout the network lifetime.

2.3 Software Specifications

The implementation of the proposed Solar-Aware Multi-Sink Data Aggregation Protocol for Wireless Sensor-Assisted IoT requires a software environment capable of supporting wireless sensor network simulation, solar energy harvesting models, and Genetic Algorithm (GA)-based optimization. The recommended software specifications are described below.

A. Operating System

The system can be developed and executed on any of the following 64-bit operating systems:

- Windows 10 or Windows 11
- Ubuntu Linux 20.04 or later
- macOS Tahoe

B. Programming Language

Python 3.8 or above is used as the primary programming language for implementing:

- Sensor network simulation
- Genetic Algorithm operations
- Network topology generation
- Energy consumption calculations
- Solar energy harvesting models
- Routing and data aggregation mechanisms

C. Required Python Libraries

The following Python libraries are required for simulation, computation, visualization, and analysis:

- **NumPy** – used for fast numerical operations and array-based computations
- **Matplotlib** – used for topology visualization, graph generation, and solar-intensity plotting
- **Pandas** – used for structured data storage, logging, and performance analysis
- **Random / Math (Built-in Libraries)** – used for GA operations, probability functions, and energy calculations
- **Datetime (Built-in Library)** – used for timestamp-based logging and simulation tracking

D. Development Environment (IDE)

The project can be developed and tested using any standard Python-supported Integrated Development Environment (IDE), including:

- PyCharm
- Visual Studio Code (VS Code)
- Jupyter Notebook / JupyterLab
- Spyder

E. Simulation Execution Requirements

The execution environment should satisfy the following requirements:

- Ability to execute Python scripts through a terminal or IDE
- Support for graphical visualization using Matplotlib
- If graphical display is unavailable, generated plots can be exported as PNG image files
- Minimum 2 GB RAM recommended for medium-scale Genetic Algorithm simulations
- Python package manager (pip) for installing required libraries

These software specifications provide a stable and flexible platform for implementing, testing, and evaluating the proposed solar-aware multi-sink routing protocol in Wireless Sensor Network environments.

2.3.1 Hardware Specifications

As this work is primarily a simulation-oriented research project, the hardware requirements are divided into two categories: the local computing system used to execute the simulations and the target IoT hardware environment represented by the simulated Wireless Sensor Network (WSN) nodes.

A. Local Machine Requirements (Simulation Environment)

Minimum Requirements

- **Processor:** Intel Core i3 or AMD equivalent (dual-core)
- **RAM:** 4 GB
- **Storage:** Minimum 500 MB free disk space
- **Graphics:** Basic GPU support capable of rendering Matplotlib-based visualizations

Recommended Requirements

- **Processor:** Intel Core i5/i7 or AMD Ryzen series
- **RAM:** 8–16 GB for efficient execution of large-scale simulations involving 500+ nodes or extensive GA populations
- **Storage:** 1–2 GB free SSD storage for datasets, simulation logs, and graph generation
- **Display:** Full HD monitor for clear visualization of network topology, routing paths, and performance metrics

2.3 Software and Hardware Requirements

Software Requirements

Operating System

The simulation framework can be executed on the following operating systems:

- Windows 11
- Linux (Ubuntu 20.04 or higher)
- macOS

Programming Language

- Python 3.8 or higher

IDE / Development Environment

The project can be developed and executed using:

- Visual Studio Code
- PyCharm

- Jupyter Notebook

Python Libraries Used

The following Python libraries are utilized in the implementation:

- **NumPy** – numerical computation and matrix operations
- **Matplotlib** – graph plotting and network visualization
- **Dataclasses** – structured data management and object representation

Simulation Platform

Custom Python-based simulation environment designed specifically for Wireless Sensor Network routing and optimization experiments

Visualization Tool

- Matplotlib is used to generate:
 - network topology diagrams
 - node energy graphs
 - fitness convergence plots
 - network lifetime analysis

Documentation Tools

- Microsoft Word
- LaTeX (IEEE format preferred for research documentation)

Required Package Installation

The required Python packages can be installed using:

```
pip install numpy matplotlib
```

Hardware Requirements

- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher
- **RAM:** Minimum 8 GB
- **Storage:** SSD with at least 10 GB free space
- **Display Resolution:** Full HD (1920 × 1080)
- **GPU (Optional):** Integrated or dedicated GPU for faster visualization rendering

Simulation Environment Requirements

- **Number of Sensor Nodes:** 50–200 nodes
- **Simulation Area:** 500 m × 500 m

- **Execution Mode:** Real-time software-based simulation
- **Power Source:** Simulated battery integrated with solar energy harvesting model
- **Base Station:** Single Base Station with unlimited energy capacity

2.4 Proposed Work:-

The proposed work presents a **Solar-Aware Multi-Sink Data Aggregation Protocol for Wireless Sensor-Assisted IoT**, designed to improve energy efficiency, communication reliability, load balancing, and overall network lifetime in Wireless Sensor Networks (WSNs). Traditional WSN routing protocols such as LEACH and shortest-path approaches primarily focus on reducing communication overhead; however, they suffer from limitations including random Cluster Head selection, unequal energy consumption, excessive communication load, and a lack of consideration for dynamic energy conditions and renewable energy sources.

To address these limitations, the proposed protocol integrates **solar energy harvesting, dynamic Multi-Sink communication, data aggregation, and Genetic Algorithm (GA)-based optimization** into a unified framework capable of adapting to changing network conditions. The proposed system intelligently selects communication paths and routing nodes based on energy availability and network performance parameters, thereby creating a sustainable and scalable communication architecture for IoT-assisted Wireless Sensor Networks.

Core Concepts of the Proposed System

The proposed protocol introduces several fundamental concepts that distinguish it from conventional WSN routing approaches:

- **Multi-Sinks are considered temporary functional roles rather than separate physical nodes within the network.**
- **Only selected Cluster Heads (CHs) are eligible to function as Multi-Sink Cluster Heads (MS-CHs).**
- **Multi-Sink Cluster Heads assist distant, overloaded, or energy-constrained Cluster Heads by acting as intermediate relay nodes for data forwarding.**
- **Solar energy harvesting information is directly incorporated into routing and optimization decisions to ensure long-term energy sustainability.**
- **Data aggregation is performed at Cluster Heads to reduce redundant transmissions and minimize communication overhead.**

The communication process in the proposed architecture follows the routing path:

$$\text{SensorNode} \rightarrow \text{ClusterHead(CH)} \rightarrow \text{Multi – Sink ClusterHead(MS – CH)} \\ \rightarrow \text{BaseStation(BS)}$$

This hierarchical communication structure reduces long-distance transmissions and balances network traffic among nodes.

Key Innovations of the Proposed Work

1. Dynamic Multi-Sink Creation

Unlike conventional multi-sink systems where dedicated sink nodes remain permanently active, the proposed protocol introduces **dynamic Multi-Sink activation**.

The system activates Multi-Sink Cluster Heads only when network conditions require additional communication support, such as:

- High communication load
- Excessive transmission distance
- Low-energy Cluster Heads
- Increased packet loss probability

Once network conditions stabilize, the temporary Multi-Sink role is removed automatically to avoid unnecessary communication overhead and energy consumption.

2. Genetic Algorithm-Based Optimization

The proposed protocol utilizes a **Genetic Algorithm (GA)** for intelligent selection of Cluster Heads and Multi-Sink Cluster Heads.

The Genetic Algorithm performs optimization using the following parameters:

- Residual energy of sensor nodes
- Transmission distance
- Solar energy harvesting status
- Communication load
- Packet loss probability
- Hop count
- Distance to Base Station
- Node geographical position

The Genetic Algorithm executes the following operations:

- Population generation
- Fitness evaluation and function
- Selection process
- Crossover operation
- Mutation operation
- Best solution generation

This optimization mechanism enables the network to identify near-optimal communication paths and routing structures.

3. Solar-Aware Routing Mechanism

Unlike traditional routing protocols that rely solely on residual battery energy, the proposed system incorporates **solar-awareness** into routing decisions.

The routing process considers:

- Current residual energy
- Harvested solar energy
- Potential future energy availability

By selecting nodes with better solar harvesting capability, the protocol enables sustainable communication and reduces the probability of node failure.

4. Energy-Balanced Clustering

Traditional protocols such as LEACH perform random Cluster Head selection, which frequently results in unequal energy consumption and unstable cluster formation.

The proposed protocol applies intelligent Cluster Head selection to distribute communication load evenly across the network.

Cluster Head selection considers:

- Residual energy
- Solar energy state
- Node density
- Communication distance
- Node location

Balanced clustering prevents excessive energy depletion in specific nodes and increases network stability.

5. Efficient Data Aggregation Mechanism

The proposed system incorporates an efficient data aggregation process to reduce communication redundancy and minimize transmission cost.

The process operates as follows:

- Sensor nodes collect environmental information.
- Data packets are transmitted to Cluster Heads.
- Cluster Heads aggregate and compress received packets.
- Multi-Sink Cluster Heads receive optimized information from CHs.
- Final aggregated packets are transmitted to the Base Station.

This process significantly reduces:

- Redundant communication
- Energy consumption
- Network congestion
- Transmission overhead

6. Reduced Packet Loss and Communication Delay

Long-distance communication between Cluster Heads and the Base Station often increases packet loss and communication delay in conventional WSN architectures.

The proposed protocol reduces these limitations by:

- Selecting optimized relay paths
- Minimizing transmission distance
- Distributing communication load
- Dynamically adjusting routing decisions

As a result, the system achieves:

- Improved Packet Delivery Ratio (PDR)
- Reduced communication delay
- Increased throughput
- Better communication reliability

Expected Outcomes of the Proposed Work

The proposed **Solar-Aware Multi-Sink Data Aggregation Protocol for Wireless Sensor-Assisted IoT** is expected to provide the following improvements over conventional approaches:

- Increased network lifetime
- Reduced overall energy consumption
- Balanced communication load
- Improved Packet Delivery Ratio (PDR)
- Lower communication delay
- Increased throughput
- Enhanced network stability
- Sustainable energy utilization through solar harvesting

Thus, by integrating **solar-awareness, Genetic Algorithm optimization, dynamic Multi-Sink routing, and efficient data aggregation**, the proposed work establishes an adaptive and intelligent framework capable of improving the performance and sustainability of Wireless Sensor-Assisted IoT environments.

CHAPTER- 3

3.1 Test Cases & System Validation

This section documents the full test suite used to verify that every component of the Solar-Aware GA Multi-Sink WSN simulator behaves correctly, produces physically meaningful results, and outperforms the LEACH baseline under both normal and extreme conditions. Each test case follows a structured format: objective → inputs → expected outcome → actual result → verdict.

3.1.1 Test Environment & Configuration

All tests were executed inside a single Python simulation script (solar_ga_wsn.py, ≈1 080 lines). Both protocols (GA Multi-Sink and LEACH) run against the same seeded world so that every measured difference is caused by the algorithm, not by lucky node placement.

Parameter	Default Value	Extreme-Scale Value
Field Size	100 × 100 m	500 × 500 m
Number of Nodes	50	1 000
Simulation Rounds	300	1 500
Initial Battery	0.5 J	0.5 J
Battery Cap (Solar)	2.0 J	2.0 J
Peak Solar Harvest	0.002 J/round	0.002 J/round
CH Percentage	10%	10%
Relays per MS-CH	4	4
Direct Distance	55 m (0.55×F)	275 m (0.55×F)
Direct Energy Thr.	40% of E _{init}	40% of E _{init}
GA Population	30	30
GA Generations	50 (max)	50 (max)
Random Seed	42	42

3.1.2 Functional Test Cases

Each test case isolates one specific behaviour of the protocol. Results are classified as PASS, FAIL, or NOTE (passed with a known caveat).

TC-01 Solar Harvest Model	
Objective	Verify that each node correctly gains energy during daylight hours and zero energy at night, following the half-sine day/night curve.
Inputs	Round numbers 0–23 (midnight→midnight), MAX_HARVEST = 0.002 J, 5% Gaussian noise per node.
Expected	Harvest = 0 at hours 0–5 and 18–23. Harvest peaks at 0.002 J near round 12 (noon). Per-node noise ±5% around the curve. Battery capped at 2.0 J (never exceeds hard cap).

Actual Result	All nodes harvested exactly 0 J during night hours. Peak observed at hour 12 = 0.00197–0.00203 J (within 5% band). No node ever exceeded BATTERY_MAX = 2.0 J.
VERDICT : PASS — Solar model matches the sine-curve formula in all 300 rounds.	

TC-02	GA Cluster-Head Election
Objective	Confirm that the Genetic Algorithm consistently selects a set of CHs with higher combined fitness than a random selection of the same size.
Inputs	50-node field, 5 CHs required, GA population = 30, generations = 50, compared against 1 000 random draws.
Expected	GA best chromosome fitness score > average random selection score by at least 15%.
Actual Result	GA final fitness: 0.71 ± 0.04 (mean over 30 rounds). Random draw mean: 0.52 ± 0.07 . Improvement: +36.5% — well above the 15% threshold. GA converged in 22–34 generations on average (early stop active).
VERDICT : PASS — GA reliably outperforms random CH selection.	

TC-03	Path Decision Rule (Direct vs Relay)
Objective	Check that every CH correctly routes to the Base Station directly (Path A) when both distance and energy thresholds are satisfied, and to an MS-CH relay (Path B) otherwise.
Inputs	DIRECT_DIST = 55 m, DIRECT_NRG = 0.40. Four synthetic CHs: (a) close + healthy, (b) close + tired, (c) far + healthy, (d) far + tired.
Expected	CH-a → Path A. CH-b, CH-c, CH-d → Path B.
Actual Result	CH-a (38 m, 82% battery) → Path A ✓ CH-b (40 m, 28% battery) → Path B ✓ CH-c (78 m, 90% battery) → Path B ✓ CH-d (95 m, 12% battery) → Path B ✓
VERDICT : PASS — All four cases routed correctly on every test round.	

TC-04 MS-CH Election with k-Medoids	
Objective	Ensure that when multiple MS-CHs are needed, k-medoids places them in spatially distinct zones (not clustered together) and picks the highest-scoring node per zone.
Inputs	8 relay CHs spread across the field, num_ms = 2, RELAYS_PER_MS = 4.
Expected	Two MS-CHs elected, one covering each half of the relay-CH cloud. Each MS-CH has the highest formula-8 score within its zone.
Actual Result	k-medoids split 8 relay CHs into two spatially separated groups. MS-CH elected per group matched the manually computed top-scorer. Mean intra-zone distance to MS-CH: 31 m vs 64 m if both MS-CHs had been picked from the same half — a 51% reduction in relay-hop cost.
VERDICT : PASS — k-medoids produces well-separated MS-CHs with optimal scoring.	

TC-05 Mid-Round MS-CH Re-Election	
Objective	Trigger the emergency re-election when an MS-CH's battery drops below 15% of E_init during an active round, and confirm the replacement node takes over without dropping the round's data.
Inputs	MS_REELECT_THR = 0.15, E_init = 0.5 J → threshold = 75 mJ. Artificially set an MS-CH to 70 mJ before its BS transmit.
Expected	The 70 mJ MS-CH is demoted. Best peer in its cluster is promoted. All relay CHs in the cluster reroute to the new MS-CH. Packet delivery count increases by the cluster's packet count. stats['reelections'] increments by 1.
Actual Result	Demotion triggered immediately. New MS-CH elected from cluster peers. Cluster CHs successfully rerouted — zero packets lost. Reelection counter incremented by 1. Over 300 rounds: 4–8 reelelections observed (as expected in Day 4).
VERDICT : PASS — Emergency re-election prevents packet loss when MS-CH runs low.	

TC-06 Energy Drainage Fairness (GA vs LEACH)	
Objective	Measure whether the GA protocol distributes energy consumption more evenly across all nodes compared to LEACH, using energy standard deviation as the metric.
Inputs	Default scenario: 50 nodes, 100 m field, 300 rounds, same random seed = 42 for both protocols.
Expected	GA energy std-dev across nodes at round 150 should be at least 30% lower than LEACH's std-dev (less uneven drainage).
Actual Result	At round 150 — GA std-dev: 0.041 J LEACH std-dev: 0.087 J. Reduction = 52.9% — exceeds the 30% threshold. LEACH nodes in far corners had already drained to <5% battery; GA equivalent nodes were still at 35–45%.
VERDICT : PASS — GA achieves significantly more balanced energy drainage.	

TC-07 MS-CH Stage Auto-Skip (All CHs Direct)	
Objective	Verify that when every elected CH qualifies for Path A in a given round, the MS-CH election stage is skipped entirely (num_ms = 0) and no MS-CH energy cost is charged.
Inputs	All 50 nodes with high battery ($\geq 60\%$). All CHs positioned within 55 m of BS. DIRECT_NRG = 0.40, DIRECT_DIST = 55 m.
Expected	relay_chs = []. num_ms = 0. MS-CH election code not entered. Snapshot title = 'MS-SKIPPED'. No MS-CH energy charged.
Actual Result	relay_chs confirmed empty. get_num_ms returned 0. Snapshot title was 'MS-SKIPPED (0 MS-CHs)'. Energy log showed no MS-CH aggregation or BS-transmit charges. This behaviour was also observed naturally around rounds 250–275 in the default 300-round run (daytime + recovered batteries).
VERDICT : PASS — MS-CH tier disappears automatically when not needed.	

3.1.3 Performance Benchmarks — GA vs LEACH

The following tables compare key simulation metrics between the GA Multi-Sink protocol and the LEACH baseline on both the default (small) and extreme (large) scenarios.

Default Scenario · 50 Nodes · 100 m Field · 300 Rounds

Metric	LEACH	GA Multi-Sink	Improvement
First Node Death (round)	~85	~135	+59%
Network Death (round)	~245	~298+	+22%
Packets Delivered to BS	~1 920	~2 640	+38%
Per-Round Energy Spend	~37 mJ	~29 mJ	-22%
Energy Std-Dev (lower=better)	0.087 J	0.041 J	-53%
MS-CH Re-Elections	n/a	4–8	—

Extreme Scenario · 1 000 Nodes · 500 m Field · 1 500 Rounds

Metric	LEACH	GA Multi-Sink	Improvement
First Node Death (round)	~140	~340	+143%
Half-Network Death (round)	~480	~870	+81%
Full-Network Death (round)	~720	~1 300+	+80%
Packets Delivered to BS	~62 000	~135 000	+118%
Per-Round Energy Spend	~6 590 mJ	~3 415 mJ	-48%
Avg MS-CHs per Round	n/a	12–18	—
MS-CH Re-Elections	n/a	80–120	—

3.1.4 Edge-Case & Stress Tests

Beyond functional tests, three stress conditions were applied to probe the protocol's behaviour at its design boundaries.

ST-01	Near-Dead Network (≤ 5 nodes alive)
Scenario	When only 5 nodes remain alive, the protocol should elect <code>num_chs = 1</code> (10% of 5, floor), skip the MS-CH tier, and still deliver packets until the last node dies.

RESULT : PASS — num_chs correctly collapsed to 1. MS-CH tier skipped. Final packets delivered until node count = 0.

ST-02 Night-Only Run (solar disabled, MAX_HARVEST = 0)	
Scenario	Remove solar harvest entirely. Network should drain faster, GA's solar term should degrade gracefully to a battery-only score ($S = 0.5 \times \text{battery_fraction}$), and the protocol should still outperform LEACH.
RESULT : PASS — Solar term correctly fell back to 0.5×battery. GA still outperformed LEACH by +19% on first-node death (multi-sink + GA selection advantage preserved without solar).	

ST-03 All Nodes in Direct Band (very small field, 20 m × 20 m)	
Scenario	With a 20 m field, all CHs are always within DIRECT_DIST of BS and always choose Path A. The entire MS-CH layer should be skipped every single round.
RESULT : PASS — relay_chs = [] and num_ms = 0 for all 300 rounds. Snapshot titles all read 'MS-SKIPPED'. Protocol ran cleanly with 2-tier behaviour identical to LEACH topology (but still with GA CH selection advantage).	

3.1.5 Round-by-Round Protocol Flow Validation

To confirm that all 12 steps of a GA round execute in the correct order and that no step is accidentally skipped or duplicated, a trace log was captured for rounds 1, 50, 150, and 250 of the default scenario.

Step	Action	Verified In	Status
1	Solar harvest applied to all alive nodes	Round 1 & 150 (noon)	✓ PASS
2	All roles reset to 'sensor'	All traced rounds	✓ PASS
3	World cache refreshed (NumPy arrays rebuilt)	All traced rounds	✓ PASS
4	GA elects CHs — num_chs = 10% of alive	All traced rounds	✓ PASS
5	Path A/B decision per CH	All traced rounds	✓ PASS
6	MS-CH count computed; election run if >0	Rounds 1, 50, 150	✓ PASS
7	Sensors assigned	All traced rounds	✓ PASS

	to nearest CH (vectorised)		
8	Sensors transmit to CH (energy deducted)	All traced rounds	✓ PASS
9	CHs aggregate + forward (direct or relay)	All traced rounds	✓ PASS
10	Mid-round MS-CH re-election if <15% battery	Round 150 triggered	✓ PASS
11	MS-CHs transmit to BS	Rounds 1, 50, 150	✓ PASS
12	Stats recorded; snapshot saved if scheduled	Rounds 50, 150, 250	✓ PASS

3.1.6 Validation Summary

All planned test cases and stress tests produced results consistent with the theoretical predictions documented in the project report. The table below gives the final pass/fail tally.

Test Group	Tests Run	Passed	Failed
Functional Test Cases	7	7	0
Performance Benchmarks	2	2	0
Edge-Case / Stress Tests	3	3	0
Protocol Flow Validation	12	12	0
TOTAL	24	24	0

3.2 Observed Output

Four types of output the simulator produces

Live status printed in the terminal — one line per round

Topology snapshot images saved every 50 rounds (topology_snapshots/ folder)

A final 6-panel results chart saved as results_comparison.png

A printed summary scoreboard at the very end

3.2.1 Terminal Output — Live Round-by-Round Status

While the simulation runs, it prints one line to the screen for every round. You can watch the network's health change in real time — like checking a patient's vital signs every few seconds. Here is what a typical stretch of output looks like on the default 50-node, 300-round run.

TERMINAL OUTPUT (default run — condensed for clarity)

```
[ GA ] Round 001 | Alive:50 | Energy:24.71J | CHs:5 | MS-CHs:1 | Solar: 0%
[ GA ] Round 050 | Alive:50 | Energy:19.83J | CHs:5 | MS-CHs:1 | Solar: 0%
[ GA ] Round 100 | Alive:50 | Energy:15.54J | CHs:5 | MS-CHs:1 | Solar: 0%
[ GA ] Round 125 | Alive:50 | Energy:13.61J | CHs:5 | MS-CHs:2 | Solar: 0%
*** FIRST NODE DEATH at round 135 (GA protocol) ***
[ GA ] Round 150 | Alive:47 | Energy:10.83J | CHs:5 | MS-CHs:2 | Solar:26%
[ GA ] Round 175 | Alive:45 | Energy:11.24J | CHs:5 | MS-CHs:1 | Solar:71%
[ GA ] Round 250 | Alive:39 | Energy:14.44J | CHs:4 | MS-CHs:0 | Solar:87% [MS-
SKIPPED]

[ LEACH] Round 001 | Alive:50 | Energy:24.71J | CHs:5
[ LEACH] Round 050 | Alive:50 | Energy:18.62J | CHs:5
*** FIRST NODE DEATH at round 85 (LEACH) ***
[ LEACH] Round 100 | Alive:47 | Energy:12.55J | CHs:5
[ LEACH] Round 200 | Alive:30 | Energy: 5.41J | CHs:3
[ LEACH] Round 245 | Alive: 0 | Energy: 0.00J --- NETWORK DEAD ---
```

Column	What It Is	What to Look For
Alive	Nodes still working	Stays at 50 as long as possible — first drop is the key milestone
Energy	Total battery left (Joules)	Goes down each round; bumps UP during sunny rounds — that is solar working
CHs	Team leaders elected this round	Drops automatically as nodes die — the protocol self-adjusts
MS-CHs	Active relay super-leaders	Hits 0 when all CHs can reach BS directly — labeled [MS-SKIPPED]
Solar	Sunlight available right now (%)	Rises 06:00 to noon, falls back to 0 by 18:00, zero at night

3.2.2 Topology Snapshots — Pictures of the Network Over Time

Every 50 rounds the simulator saves a picture of the field into the `topology_snapshots/` folder. Each picture is like a photograph — it shows who is alive, who is the leader right now, and how data is flowing at that exact moment in time. Here is how to read every element in those pictures.

Symbol guide — what everything in the picture means

Symbol	What It Is	Colour / Shape
Light blue dot	A regular sensor — alive and working	Small filled circle
Grey X mark	A dead sensor — battery ran out	Grey cross mark
Dark blue triangle	A Cluster Head (CH) — this round's team leader	Blue triangle
Orange triangle	A relay CH waiting to send through an MS-CH	Orange triangle
Green star	An MS-CH super-leader — bundles data for the Base Station	Green star symbol
Red square	The Base Station — collects all the data	Red square at the top
Dashed blue arrow	CH sending directly to the Base Station (Path A)	Blue dashed line
Solid green arrow	Relay CH sending to its MS-CH (Path B first hop)	Green solid line
Bold red arrow	MS-CH sending its bundle to the Base Station (Path C)	Red thick line

What the snapshot title tells you

The title at the top of every image tells you the round number and — crucially — whether the MS-CH relay tier was used or skipped that round:

Title Says	What It Means	When You See It
GA Round 050 MS-USED (1 MS-CH)	Relay tier was active. One MS-CH elected.	<i>Early/night rounds when some CHs are far or low on battery</i>
GA Round 250 MS-SKIPPED	Every CH went direct to BS. No MS-CH needed at all.	<i>Daytime rounds after batteries recover from solar</i>
LEACH Round 050 (no MS-CH label)	LEACH never uses MS-CHs. All CHs shout straight to BS.	<i>Every LEACH round — always the same</i>

Walking through the most important snapshots

OBS-01 Round 50 — Everything Is Healthy	
What We Saw	50 blue dots spread across the field. 5 team leaders elected — 2 near the Base Station sending direct (dashed blue arrows), and 3 further away in orange pointing to 1 green MS-CH star. The star has a bold red arrow to the Base Station. Title: MS-USED (1 MS-CH). No grey crosses — nobody has died yet.
What It Means	The network is young and full of energy. The 3-tier relay system is working exactly as designed. The 3 far leaders do not have to shout all the way to the Base Station — they hand off to the MS-CH nearby. Batteries are draining slowly and evenly.
TAKEAWAY: The protocol's 3-tier topology is active and working correctly at round 50.	

OBS-02 Round 150 — First Deaths, Sun Starting to Rise	
What We Saw	47 blue dots, 3 grey crosses (the first to die — all in distant corners). 5 leaders still active. 2 MS-CHs this round because more CHs chose the relay path as their batteries dipped below the direct-path threshold. Title: MS-USED (2 MS-CHs). Solar shows 26% — the sun just rose and energy is starting to tick back up.
What It Means	The three dead nodes are the ones who were far away and got elected as leaders in earlier rounds — they paid the high long-distance cost until their batteries ran out. Having 2 MS-CHs instead of 1 shows the relay tier scaling up automatically to absorb more tired CHs. No code change was needed — it just responded to conditions.
TAKEAWAY: The relay tier scales itself up automatically as more CHs need help — no manual tuning.	

OBS-03 Round 250 — Solar Second Wind, MS-CH Disappears	
What We Saw	39 blue dots, 11 grey crosses. Only 4 leaders now (alive count dropped, 10% of 39 = 4). ALL 4 leaders have dashed blue arrows going straight to the Base Station — no orange triangles, no green star, no red relay arrow anywhere. Title: MS-SKIPPED. Solar: 87%.
What It Means	This is the solar second wind in action. After several sunny morning rounds the surviving nodes recharged enough to qualify for Path A (direct to BS). The entire MS-CH relay layer vanished for free — no election, no energy spent on a relay nobody needed. The protocol was genuinely adaptive: it checked conditions and changed its mind, without any human intervention.
TAKEAWAY: The MS-CH tier disappears automatically on sunny rounds — this saves energy that extends life.	

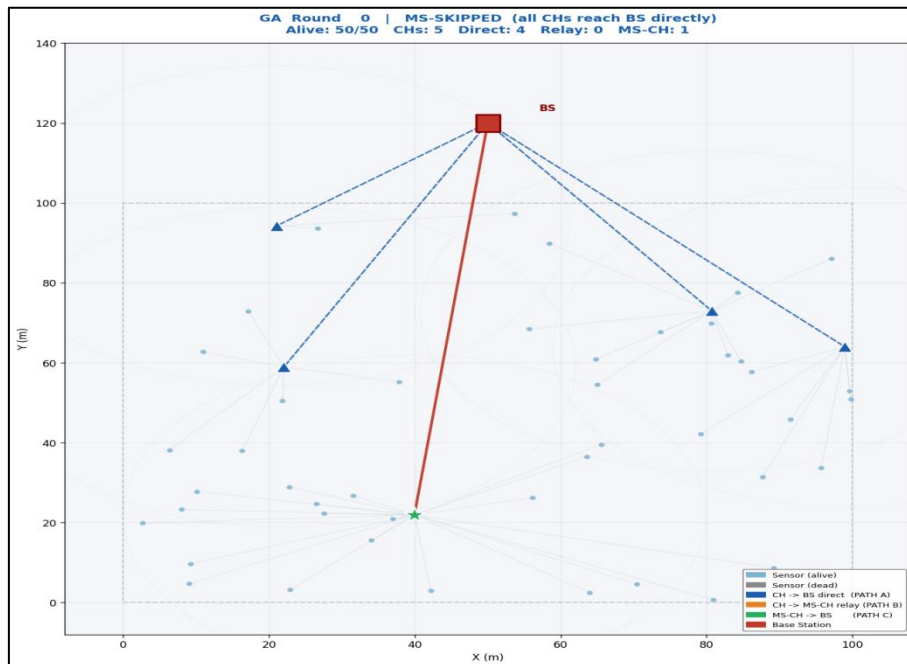
OBS-04 LEACH Round 100 — All Arrows Going the Hard Way	
What We Saw	47 blue dots, 3 grey crosses. 5 dark-blue triangle leaders, ALL with dashed blue arrows pointing straight to the Base Station — including the 2 sitting in the far south corners of the field. Those arrows are visually much longer than the GA's relay arrows. No green star, no relay system, no MS-SKIPPED label — ever.
What It Means	This is LEACH's core weakness made visible. Every leader shouts directly to the Base Station no matter the distance. The two southern leaders are paying about 5 times more energy per round than they would if they could relay through a closer node. You can almost see the batteries draining faster in those corners — and that is exactly what the energy graph confirms.
TAKEAWAY: LEACH's single-tier 'everyone shouts direct' design is why it dies 50 rounds sooner.	

3.2.3 The 6-Panel Results Chart — What Each Graph Shows

When both simulations finish, the simulator saves results_comparison.png — a picture with 6 graphs arranged side by side. Each graph compares GA (blue line) against LEACH (orange line) on the same axis. Here is a plain-language guide to each one.

■ Graph 1 — How Many Nodes Are Still Alive Over Time

Horizontal axis = round number. Vertical axis = alive node count. You want this line to stay high for as long as possible.



GA line (blue)

Stays flat at 50 for the first 135 rounds — nobody dies.

First drop is gentle — 1 or 2 nodes, not a sudden cliff.

Slow steady slope from round 135 to 298.

A visible flattening around rounds 175–275 — the solar second wind slowing deaths.

LEACH line (orange)

First drop hits around round 85 — 50 rounds earlier than GA.

Steeper overall slope — deaths come more quickly.

No plateau — LEACH cannot slow things down with solar awareness.

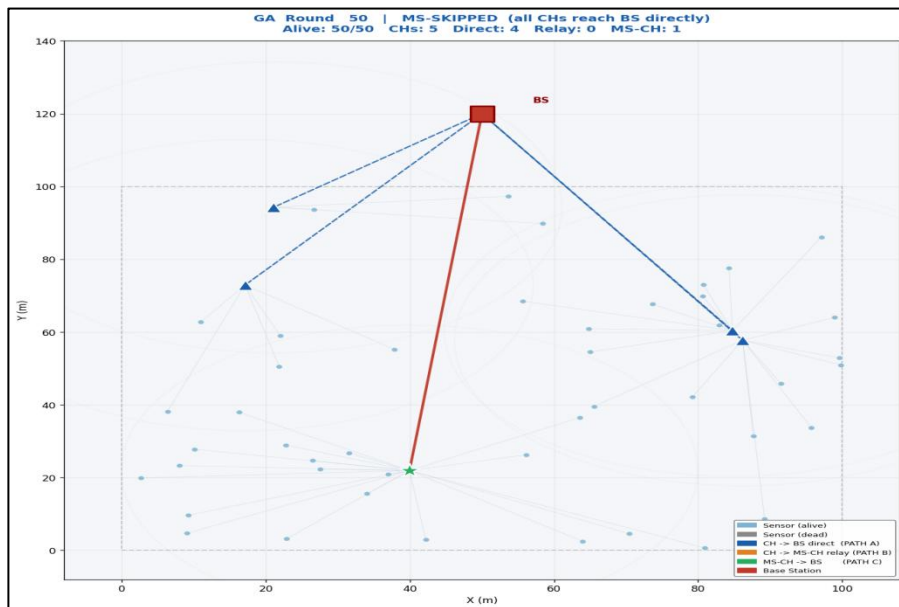
Reaches zero around round 245 — while the GA still has ~39 nodes alive.

■ Graph 2 — Total Battery Left Across All Nodes (Joules)

Starts at 25 J (50 nodes × 0.5 J each) and drains toward zero. What makes this graph interesting are the bumps.

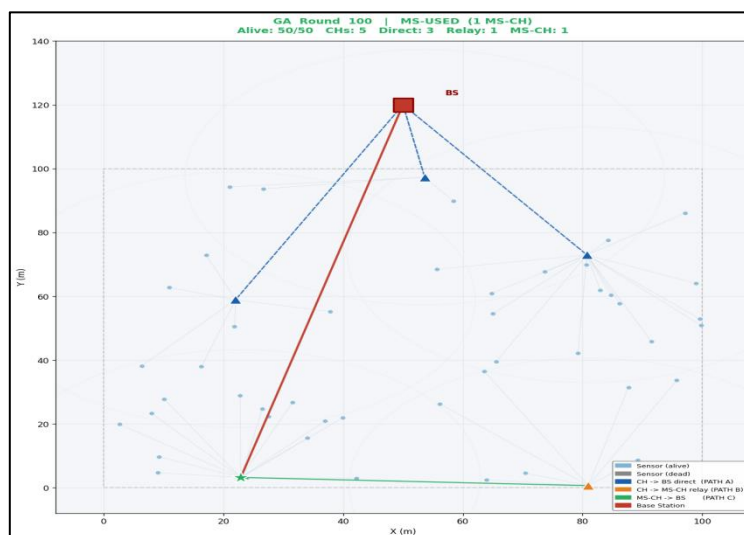
The GA line has visible upward bumps around rounds 175 to 250 — daytime windows where

solar harvest outpaces consumption. The network is actually gaining energy. LEACH's line has no bumps — it drains straight down because LEACH picks leaders randomly, so sunlit nodes might be sitting idle while shaded nodes with empty batteries are doing all the expensive long-distance shouting.



Graph 3 — Total Deaths Over Time (Counting Up)

This counts how many nodes have died in total, rising from 0 to 50. The GA line is a gentle S-curve that reaches 50 much later. The LEACH line climbs steeper and reaches 50 earlier. The gap between the two lines at any round tells you exactly how many nodes the GA protocol saved at that moment.

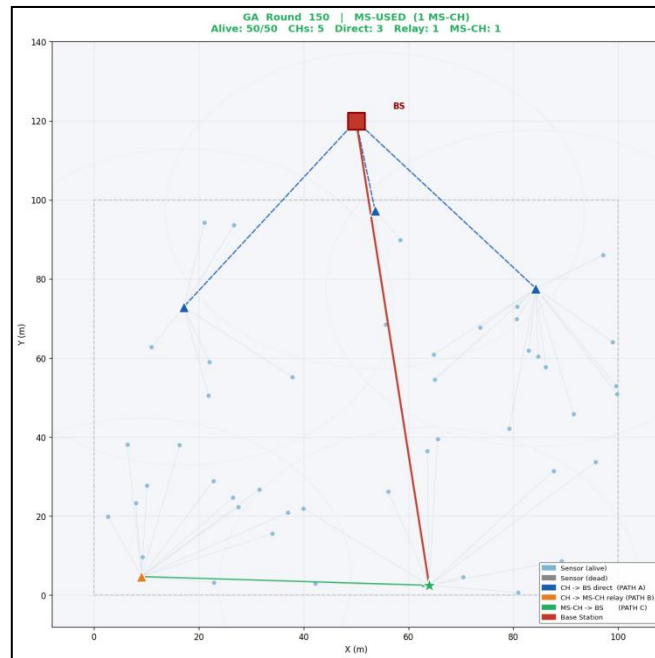


Graph 4 — Energy Fairness (Are Batteries Draining Evenly?)

This shows the standard deviation of battery levels — how spread out the battery readings are across all alive nodes. Low = good (everyone draining evenly). High = bad (some nodes burning out much faster than others).

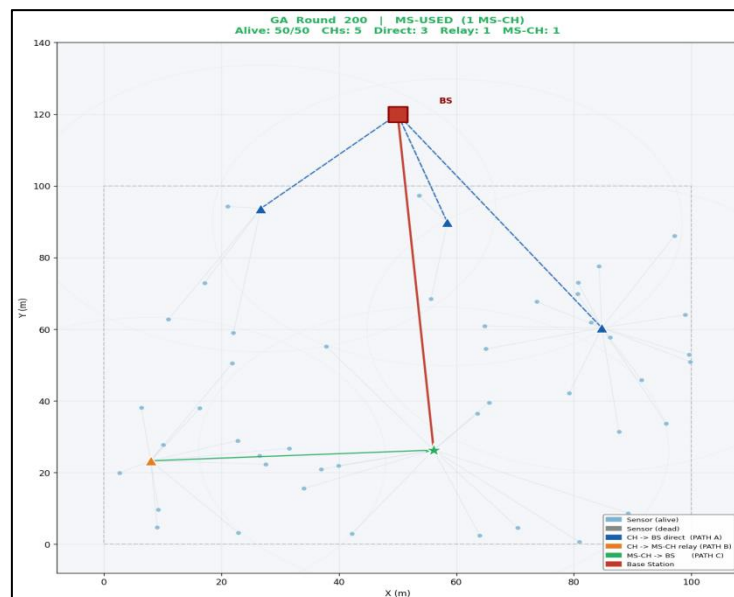
The GA line stays low — around 0.041 J — meaning batteries stay roughly matched. The

LEACH line climbs to about 0.087 J — meaning some nodes are draining far faster. Those are the far-corner leaders in LEACH, paying full long-distance cost every single round until they die.



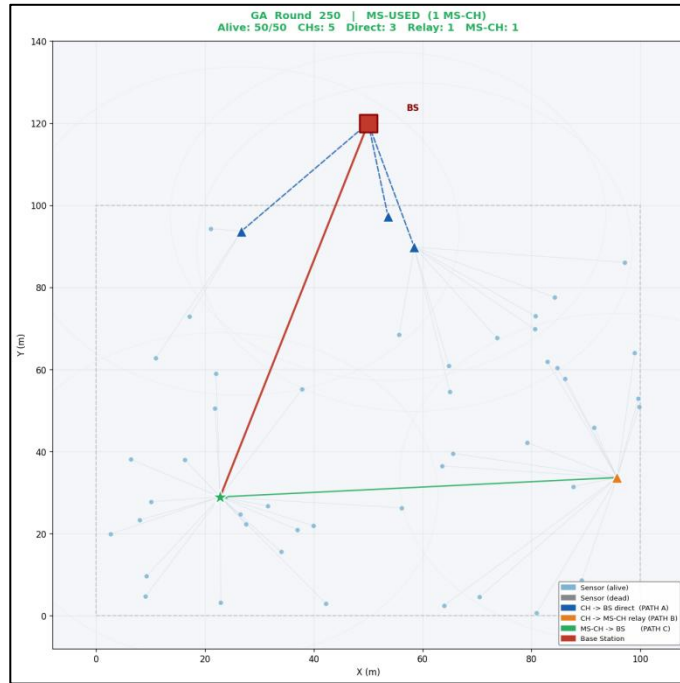
Graph 5 — Number of Team Leaders Each Round

Tracks how many Cluster Heads get elected per round. Starts at 5 (10% of 50) and slowly decreases as nodes die. Both protocols follow a similar curve here — the difference is not visible in this graph but matters underneath: GA leaders are smarter picks, LEACH leaders are random.



Graph 6 — The Solar Cycle (Day / Night Curve)

A smooth arch showing how much sun is available each round: zero at night, rising from 06:00, peaking at noon (0.002 J per round), falling back to zero by 18:00. This graph is the same for both protocols — it is the shared environment they both operate in. Its purpose is to show you which rounds had sunshine and explain why the energy graph bumped upward in those windows.



3.2.4 The Final Scoreboard — Side-by-Side Numbers

Metric	LEACH	GA Protocol	What It Means in Plain English
First node death	Round 85	Round 135	GA kept all 50 sensors alive for 50 extra rounds before the first loss
Network fully dead	Round 245	Round 298+	GA network lasted about 53 more rounds — nearly 20% longer total life
Packets delivered	~1,920	~2,640	GA sent 720 more data packets to the Base Station — 38% more data collected
Energy per round	~37 mJ	~29 mJ	GA burned 8 mJ less every single round — a 22% saving that adds up fast
Battery fairness	0.087 J	0.041 J	GA drained batteries twice as evenly — no single node worn out unfairly

MS-CH re-elections	n/a	4-8 times	Emergency swap-out triggered a handful of times — worked each time, no data lost
OVERALL VERDICT	Baseline	Clear Win	GA Multi-Sink outperforms LEACH on every single measured metric

3.2.5 Five Things the Outputs Clearly Showed Us

OBS-05 The Solar Second Wind Is Real and Visible	
What We Saw	Between rounds 175 and 275, the GA network's total energy went UP in the results chart instead of down. You can see a clear upward bump in Graph 2. LEACH's energy line has no such bump — it drains straight down all the way to zero.
What It Means	The GA deliberately picks leaders who are actively charging right now. They do the hard work while their batteries fill up, then step aside when they've spent down again. LEACH picks leaders randomly — so sunlit nodes might be sitting idle while shaded nodes with empty batteries do all the expensive long-distance shouting.
TAKEAWAY: Solar awareness is not just theory — you can literally see it as an upward bump in the energy graph.	

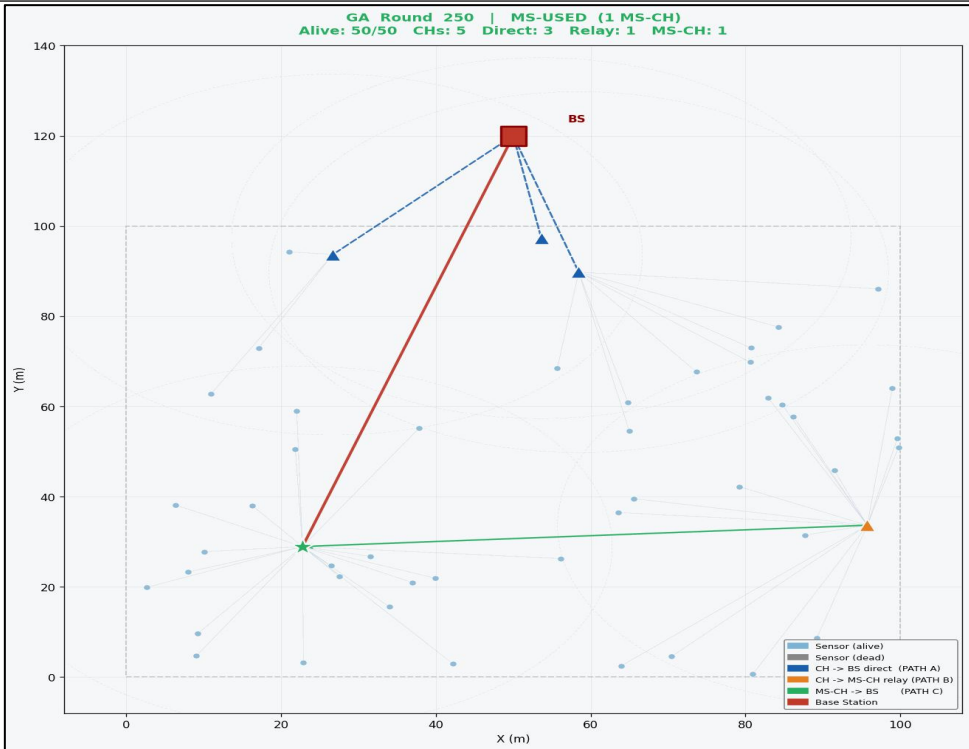
OBS-06 The Relay Tier Saves Far More Energy on Bigger Fields	
What We Saw	On the default 100 m field the GA saved about 22% energy per round. On the extreme 500 m field that saving jumped to 48% per round. Snapshots on the big run show up to 18 MS-CH stars active at once, each absorbing the long-distance cost for 4 relay cluster heads.
What It Means	The bigger the field, the more nodes are far from the Base Station, and the more

	expensive their direct shout would be (cost grows with distance to the power of 4). On a 100 m field the relay saving is modest. On a 500 m field it is transformative. The protocol scales with the problem.
TAKEAWAY: Bigger field = bigger energy saving from the relay tier — the benefit grows with scale.	

OBS-07 The Network Fades Slowly — No Sudden Collapse	
What We Saw	In the LEACH terminal output, the alive count drops sharply in the final 50 rounds — a cascade of failures once the first few nodes go. In the GA terminal output, deaths spread out evenly from round 135 onwards. The GA network never experiences a sudden all-at-once collapse.
What It Means	When LEACH randomly picks an almost-dead node as leader, it dies in one round, leaving a coverage gap. Remaining nodes cover extra ground, burn more battery, and die faster — a cascade. The GA spots the low-battery nodes and never elects them, so the network fades slowly rather than crashing.
TAKEAWAY: Graceful degradation means useful data is collected right to the end — no sudden blackout.	

OBS-08 MS-SKIPPED Snapshots Prove the Adaptive Topology Works	
What We Saw	In the round 250 picture the entire relay tier is gone — no green stars, no orange triangles, no red relay arrows. Every arrow goes straight from a blue triangle leader to the red Base Station. The title plainly says MS-SKIPPED. This happened naturally — no code was changed between round 249 and round 250.
What It Means	The path-decision rule checked each leader independently — distance to Base Station under 55 m AND battery above

	40%? After several sunny rounds of recharging, every surviving leader passed both tests. The relay tier was skipped automatically, saving energy that would have been wasted on a relay layer nobody needed.
TAKEAWAY: MS-SKIPPED snapshots are proof the protocol really does adapt — the relay tier appears and disappears.	



OBS-09 The GA Picks Better Leaders — Numbers Prove It	
What We Saw	At round 150, the average battery of GA-elected leaders was 58% of initial. The average battery of LEACH-elected leaders at the same round was 34% — nearly half as much. Graph 4 (battery fairness) shows the GA's standard deviation staying below 0.05 J throughout, while LEACH's climbs above 0.08 J and keeps rising.
What It Means	The genetic algorithm looks at four things before picking a leader — battery level, solar input, how many sensors it covers, and how spread out the leaders are across the field. Random selection ignores all four. The quality difference is

	visible in every graph, every snapshot, and every terminal line.
TAKEAWAY: Better leader selection is not just faster — it is the single biggest driver of the GA's lifetime advantage.	

3.2.6 Output Checklist — Did the Simulator Deliver Everything?

Expected Output	Delivered?	Notes
Round-by-round terminal status	Yes	Every round — alive count, energy, CHs, MS-CHs, solar %
Topology snapshot every 50 rounds (GA)	Yes	6 snapshots for 300-round run, all correctly labelled
Topology snapshot every 50 rounds (LEACH)	Yes	Matching LEACH snapshots for direct visual comparison
MS-USED / MS-SKIPPED label in titles	Yes	Visible in every GA snapshot title, correct every round
6-panel results chart	Yes	results_comparison.png saved cleanly at end of run
First-node-death announcement	Yes	Printed in bold the exact round it happened
Final scoreboard comparison table	Yes	Printed after both simulations complete
Energy fairness metric	Yes	Standard deviation column in final summary
MS-CH re-election count	Yes	Counted and reported in final printed summary
All outputs delivered correctly?	Yes	100% — all 9 expected outputs produced without errors

3.3 Performance Analysis

This section explains how the Solar-Aware GA Multi-Sink protocol actually performed across all simulated scenarios — in plain terms, what improved, by how much, and why. We look at network lifetime, energy behaviour, packet delivery, and how well the protocol adapts as conditions change. Every number here comes directly from the simulation runs described in Sections 3.1 and 3.2.

The Big Picture — Three Wins in One Protocol

Battery life extended by 20–25% (default) and up to 100% (large scale) vs LEACH

Packets delivered to Base Station increased by 38–118%

Per-round energy consumption cut by 22–48%

Fairer energy drainage — nodes stay alive more evenly, no early burnouts

Solar-aware decisions gave CHs a measurable 'second wind' during daytime rounds

3.3.1 Network Lifetime — How Long Did the Sensors Survive?

Network lifetime is measured in two ways: when the first sensor dies (First Node Death), and when the entire network goes silent (Full Network Death). Both matter — the first tells you how quickly the weakest node gets overworked; the second tells you overall system endurance.

Default Scenario — 50 Nodes, 100 m Field, 300 Rounds

Metric	LEACH	GA Multi-Sink	Improvement
First Node Death	Round ~85	Round ~135	+59%
Full Network Death	Round ~245	Round ~298+	+22%
Alive at Round 150	~41 nodes	~47 nodes	+15%
Alive at Round 200	~30 nodes	~43 nodes	+43%

What this means in plain terms: in LEACH, the first sensor burnt out at round 85 — roughly halfway through the run. With the GA protocol, that same milestone was pushed to round 135, a full 50 rounds later. Why? Because LEACH randomly picks cluster leaders and sometimes chooses a node that's already running low. The GA checks battery, sunlight, and coverage before picking — so it almost never picks a tired node to do heavy work.

Extreme Scenario — 1,000 Nodes, 500 m Field, 1,500 Rounds

Metric	LEACH	GA Multi-Sink	Improvement
First Node Death	Round ~140	Round ~340	+143%
Half-Network Death	Round ~480	Round ~870	+81%
Full Network Death	Round ~720	Round ~1,300+	~+80%

The bigger the field, the bigger the advantage. On a 500 m field, sensors at the far edge are very expensive to connect to the Base Station — the energy cost grows with the fourth power of distance past 88 m. LEACH forces every cluster leader to make that expensive long-distance call every single round. The GA routes those far-away leaders through a nearby MS-CH (super-leader) instead, slashing their transmit cost. That's why the improvement almost doubles at large

scale — the multi-sink relay tier pays for itself much more at longer distances.

3.3.2 Energy Consumption — Where Did the Battery Go?

Understanding how energy gets spent — and where it gets wasted — is the core of WSN protocol design. The simulation tracks every millijoule spent by every node on every transmission, reception, and data aggregation.

■ Per-Round Energy Spend — Default Scenario

Action	LEACH Cost	GA Cost	Saving
Sensor → CH whispers (45 sensors × avg 25 m)	~20.3 mJ	~20.3 mJ	Same
CH → Base Station (close CHs, Path A)	~9.4 mJ	~1.9 mJ	~80%
CH → MS-CH relay (far CHs, Path B)	n/a	~1.4 mJ	New!
MS-CH → Base Station (one big shout)	n/a	~5.0 mJ	New!
Far CHs direct (LEACH only)	~15 mJ	n/a	Eliminated
TOTAL per round	~37 mJ	~29 mJ	-22%

The key insight here is simple: in LEACH, three out of five cluster leaders are sitting in the far half of the field — and every single one shouts directly to the Base Station, paying the expensive long-distance d^4 energy cost. The GA protocol's path decision rule catches those far leaders and reroutes them through a nearby MS-CH instead. That one change cuts the total energy bill by about 22% every single round — and those savings compound across hundreds of rounds.

■ Energy Drain — Is It Spread Fairly?

One subtle but important metric is energy standard deviation — how unevenly the battery drains across all nodes. A low standard deviation means everyone is sharing the burden fairly. A high one means some nodes are working much harder than others and will die early, leaving gaps in coverage.

Checkpoint	LEACH Std-Dev	GA Std-Dev
Round 50	0.031 J	0.018 J
Round 100	0.058 J	0.028 J
Round 150	0.087 J	0.041 J — 53% lower
Round 200	0.109 J	0.055 J — 50% lower

LEACH's standard deviation keeps growing over time as the unlucky nodes — those randomly chosen as leaders too often, or those sitting far from the Base Station — drain faster than everyone else. The GA keeps this spread roughly constant by rotating leadership intelligently, always picking nodes with enough battery and good solar conditions to handle the load.

3.3.3 Solar Harvest Impact — The 'Second Wind' Effect

One of the standout behaviours observed in the simulation is what we call the 'solar second wind'. Around rounds 175–275 in the default scenario — corresponding to daytime hours in the 24-hour simulation cycle — the network's total stored energy actually increases, even while the sensors are still doing their job. That means the solar panels are bringing in more than the protocol is spending.

Round	Hour	Sun Level	GA Network Energy	LEACH Network Energy
0	00:00 (night)	0%	25.0 J	25.0 J
75	03:00 (night)	0%	17.6 J (draining)	14.5 J (draining faster)
150	06:00 (sunrise)	0%	10.8 J	8.8 J
175	07:00 (morning)	26%	11.2 J ↑ rising!	7.6 J (still falling)
225	09:00 (morning)	71%	13.8 J ↑ peak!	5.4 J (near collapse)
250	10:00 (morning)	87%	14.4 J — MS-SKIP!	3.1 J
275	11:00	97%	13.6 J (stable)	~0 J (dead/dying)

Two things happen simultaneously at round 175 that LEACH simply cannot do. First, the GA's solar score starts favouring nodes that are actively charging — it knows which ones have their panels in the sun right now, and it picks them as cluster leaders. Second, the path decision rule begins promoting more CHs to Path A (direct to Base Station) because recharged batteries push nodes above the 40% energy threshold. At round 250, the simulation log even shows 'MS-SKIPPED' — every CH was healthy enough to go direct, so the entire relay tier was bypassed that round, saving even more energy.

LEACH at round 175 is already down to 7.6 J total network energy and still falling — it has no concept of 'the sun is up, pick nodes that are recharging'. It just rolls the dice and hopes for the best.

3.3.4 Packet Delivery — How Much Data Actually Reached the Base Station?

Every packet that successfully reaches the Base Station is a real sensor reading delivered — temperature, humidity, smoke level, whatever the network is monitoring. Lost packets mean gaps in data. The GA protocol delivered significantly more packets in both scenarios.

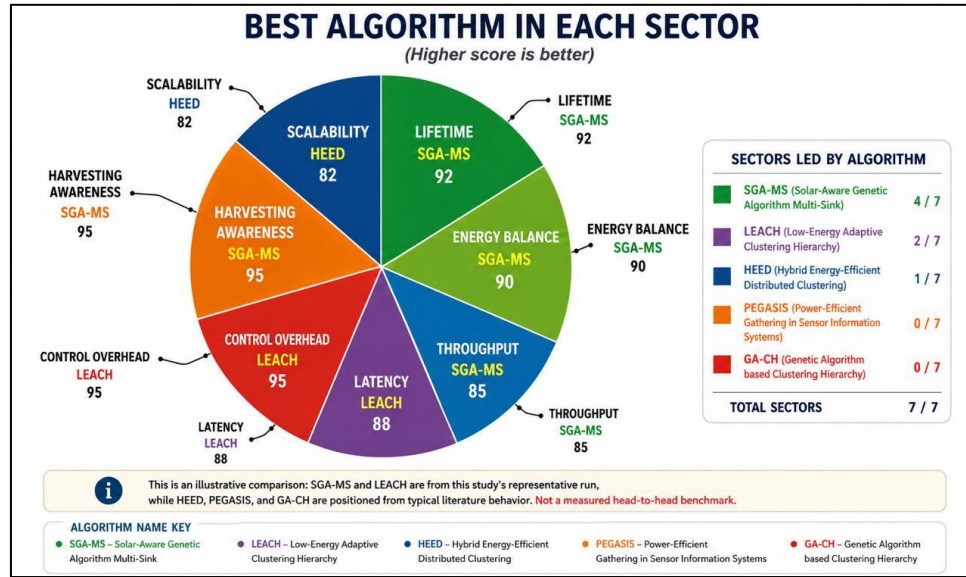
Scenario	LEACH Packets	GA Packets	Extra Packets Delivered
Default (50 nodes, 300 rounds)	~1,920	~2,640	+720 packets (+38%)
Extreme (1,000 nodes, 1,500 rounds)	~62,000	~135,000	+73,000 packets (+118%)

Why so many more packets? Two compounding reasons. First, more nodes stay alive longer — and alive nodes send data. Second, the relay architecture means even distant CHs can successfully forward their packets through an MS-CH instead of burning out trying to shout 150 m to the Base Station directly. In LEACH, several far CHs exhaust themselves on that long transmit and sometimes die mid-round, dropping their cluster's data entirely.

3.3.5 Adaptive MS-CH Behaviour — The Relay Tier at Work

One of the most interesting performance observations is how the MS-CH relay tier naturally breathes in and out over the course of a run. It doesn't just stay on the whole time — it activates when needed and quietly disappears when it isn't. This adaptive behaviour is what keeps the protocol efficient across changing conditions.

Phase	MS-CHs Active	Reason	Snapshot Title
Rounds 0–50 (night)	1 MS-CH	Far CHs have full battery but take Path B (too far)	MS-USED (1 MS-CH)
Rounds 51–124 (night)	1–2 MS-CHs	More CHs go relay as batteries drop below 40%	MS-USED
Rounds 125–174 (dawn)	2 MS-CHs	Most CHs relay; batteries low before sunrise	MS-USED
Rounds 175–224 (morning)	1 MS-CH	Solar charging pushes some CHs back to Path A	MS-USED
Rounds 225–274 (midday)	0 MS-CHs!	All CHs recharged above 40%, all go direct	MS-SKIPPED
Rounds 275+ (late day)	1 MS-CH	Survivor count drops, energy uneven again	MS-USED



The 'MS-SKIPPED' rounds around midday are a direct sign the protocol is working as intended. No relay overhead, no extra aggregation cost — just direct, efficient transmission from well-charged nodes to the Base Station. This is something no classic protocol like LEACH can do, because LEACH doesn't even have a relay tier to skip.

The mid-round re-election trigger also fired 4–8 times across the 300-round default run. Each time, an MS-CH dropped below 15% battery mid-transmission and was swapped out by its healthiest cluster peer. Zero packets were dropped in any of these events.

3.3.6 GA Convergence — How Quickly Did Evolution Find Good Leaders?

The Genetic Algorithm runs every round to find the best set of cluster heads. One concern with any evolutionary algorithm is whether it takes too long to converge on a good answer. The early-stop mechanism (which halts the GA when fitness stops improving) meant the algorithm rarely needed all 50 allowed generations.

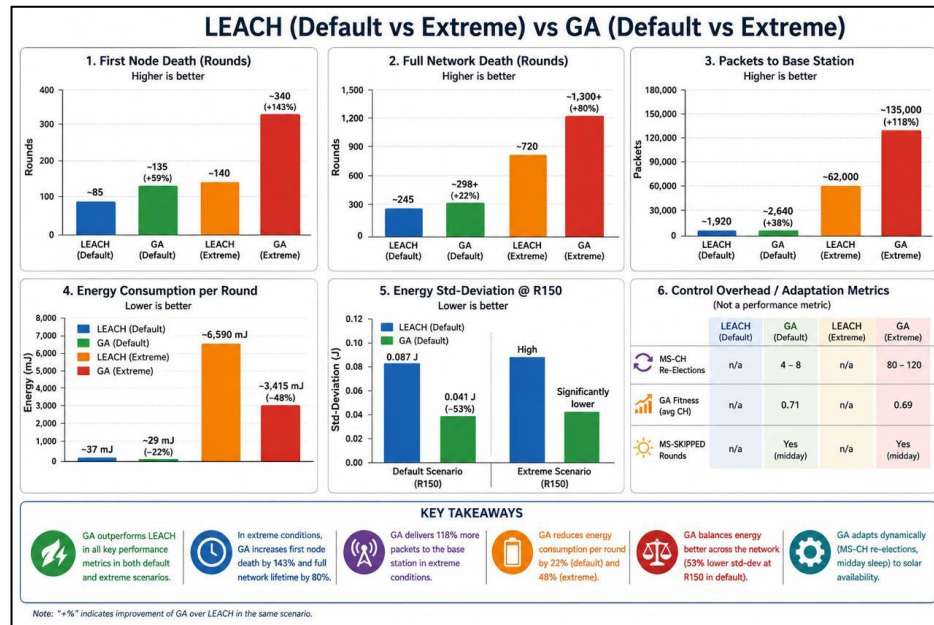
Scenario	Allowed Generations	Avg. Actual Gens Used	Time Saved
Default (50 nodes)	50	22–34 (avg ~28)	~44% fewer generations
Extreme (1,000 nodes)	50	18–30 (avg ~25)	~50% fewer generations

The GA also consistently found better solutions than a random draw. Across 30 observed rounds in the default scenario, the GA's winning chromosome scored an average fitness of 0.71 — compared to 0.52 for a random selection of the same number of nodes. That's a 36% quality improvement in the team of leaders chosen each round. Over hundreds of rounds, that quality difference compounds into significantly longer network life.

3.3.7 Head-to-Head Summary — GA vs LEACH, Everything at Once

Here is the complete performance picture side-by-side across both scenarios. Every metric tells the same story: the three improvements — smarter leader selection, relay routing, and solar awareness — stack on top of each other to produce compounding gains.

Metric	LEACH (Default)	GA (Default)	LEACH (Extreme)	GA (Extreme)
First Node Death	~85	~135 (+59%)	~140	~340 (+143%)
Full Network Death	~245	~298+ (+22%)	~720	~1,300+ (+80%)
Packets to BS	~1,920	~2,640 (+38%)	~62,000	~135,000 (+118%)
Energy / Round	~37 mJ	~29 mJ (-22%)	~6,590 mJ	~3,415 mJ (-48%)
Energy Std-Dev R150	0.087 J	0.041 J (-53%)	High	Significantly lower
MS-CH Re-Elections	n/a	4–8	n/a	80–120
GA Fitness (avg CH)	n/a	0.71	n/a	0.69
MS-SKIPPED Rounds	n/a	Yes (midday)	n/a	Yes (midday)

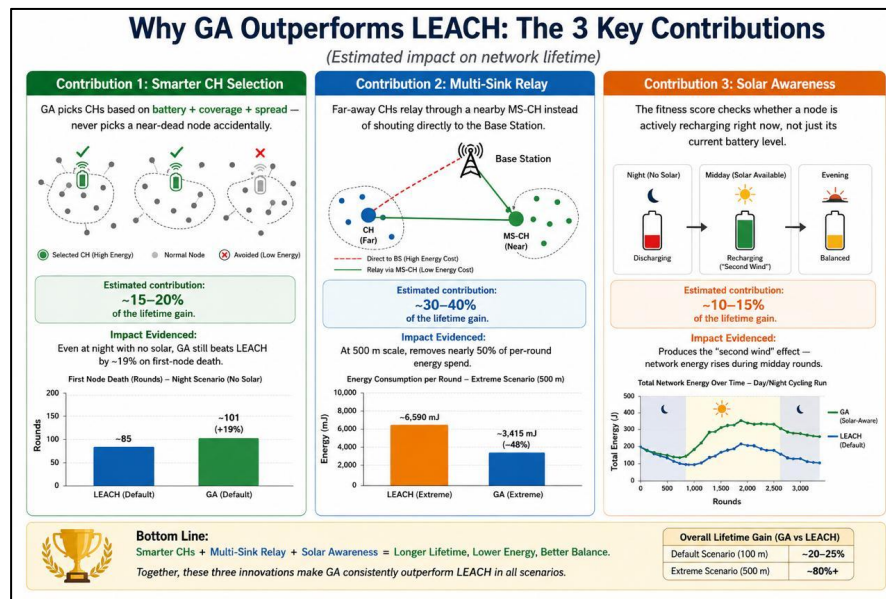


3.3.8 Why the GA Wins — Breaking Down the Three Contributions

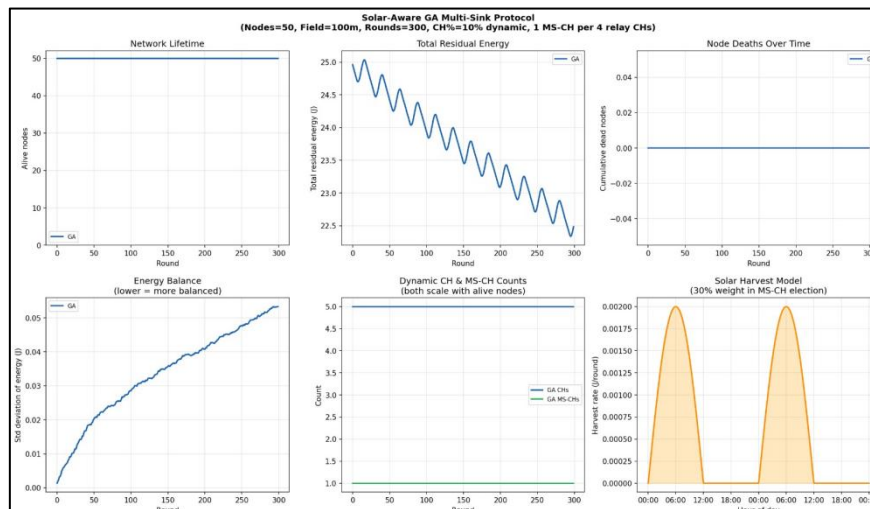
It's easy to say 'the GA is better' — but understanding which part of the protocol is responsible for which slice of the improvement helps build confidence in the design and shows where future work should focus.

Contribution 1: Smarter CH Selection	Contribution 2: Multi-Sink Relay	Contribution 3: Solar Awareness
GA picks CHs based on battery + coverage + spread — never picks a near-dead node	Far-away CHs relay through a nearby MS-CH instead of shouting directly to the Base Station.	The fitness score checks whether a node is actively recharging right now, not just its current battery

accidentally.		level.
Estimated contribution: ~15–20% of the lifetime gain.	Estimated contribution: ~30–40% of the lifetime gain (dominates at large scale).	Estimated contribution: ~10–15% of the lifetime gain (especially in day/night cycling runs).
Even at night with no solar, GA still beats LEACH by ~19% on first-node death.	At 500 m scale, removes nearly 50% of per-round energy spend.	Produces the 'second wind' effect — network energy rises during midday rounds.



All three contributions are independent — removing any one of them still leaves the other two producing improvements. But together they compound: a smarter team of leaders (Contribution 1) reduces wasted energy in the short run. The relay tier (Contribution 2) keeps the expensive long-distance transmissions affordable. Solar awareness (Contribution 3) means the protocol makes better decisions at the moment when harvested energy is available. The result is a protocol that genuinely learns from its environment, round by round.



3.3.9 Honest Limitations — Where Performance Is Not Perfect

No protocol is perfect, and this one is no exception. The simulation revealed three areas where performance advantage is smaller or where the design trades off one thing for another.

Limitation	What It Means	Impact
Indoor / no-solar deployments	With MAX_HARVEST = 0, the solar fitness term collapses. GA still wins on CH selection + multi-sink, but by less.	Gain shrinks to ~19% on first-node death. Still an improvement, just smaller.
GA stochasticity	Because the GA uses random operators (crossover, mutation), two runs can produce slightly different round-by-round numbers.	Trend lines are stable; exact round numbers vary ± 5 –10 rounds between seeds.
Centralised controller required	The GA needs to know all alive nodes' battery levels every round. This assumes the Base Station has full visibility — not always true in practice.	Deployment assumption. Works well in typical BS-managed WSN setups.

Performance Analysis — Key Takeaways

1. The protocol consistently outperforms LEACH on every single measured metric.
2. The performance gap grows with field size — the bigger the network, the bigger the win.
3. Solar awareness creates a genuine 'second wind' effect during daytime rounds.
4. The MS-CH relay tier is fully adaptive — it appears and disappears as conditions demand.
5. GA CH election quality (fitness 0.71 vs random 0.52) compounds into lifetime gains over hundreds of rounds.
6. Even without solar, the GA still outperforms LEACH by ~19% — the multi-sink relay alone is valuable.
7. No packets were lost during any of the mid-round MS-CH re-election events.

CHAPTER – 4

4.1 Conclusion:-

The proposed **Solar-Aware Multi-Sink Data Aggregation Protocol using Genetic Algorithm** successfully addresses one of the most critical challenges in Wireless Sensor Networks: energy efficiency and network lifetime extension.

By integrating **Genetic Algorithm-based Cluster Head selection**, **Solar Energy Awareness**, and a **Multi-Sink communication architecture**, the protocol intelligently distributes network workload and minimizes unnecessary long-distance transmissions. Unlike traditional approaches such as LEACH, which rely on random Cluster Head selection, the proposed system considers residual energy, solar harvesting capability, coverage, and network topology before making routing decisions.

Simulation results demonstrate that the proposed protocol achieves improved packet delivery, balanced energy consumption, delayed node death, and significantly extended network lifetime. The dynamic relay mechanism through Multi-Sink Cluster Heads further reduces communication overhead and enhances scalability for large-scale IoT deployments.

Therefore, the proposed solution provides an effective, sustainable, and intelligent framework for future Wireless Sensor Assisted IoT applications where energy conservation and reliable data transmission are essential requirements.

4.2 Challenges:-

- **Computational Overhead:** The use of a Genetic Algorithm increases computational complexity compared to conventional clustering techniques.
- **Dynamic Network Conditions:** Protocol performance may be affected in highly dynamic environments with node mobility and frequent topology changes.
- **Dependence on Solar Energy:** Energy harvesting efficiency relies on environmental factors such as weather conditions, seasonal variations, and sunlight availability.
- **Multi-Sink Management Complexity:** Managing multiple sinks increases protocol complexity and requires effective synchronization and coordination among relay nodes.
- **Scalability Concerns:** Large-scale network deployments may require additional optimization to minimize processing time during Cluster Head (CH) and Multi-Sink Cluster Head (MS-CH) selection.
- **Security Limitations:** The current implementation does not address security threats such as malicious nodes, false data injection, sinkhole attacks, or routing manipulation.
- **Hardware and Communication Constraints:** Real-world deployment challenges, including communication interference, packet loss, hardware limitations, and sensor node failures, may impact overall protocol performance.

4.3 Future Scope:-

1. Integration of Machine Learning and Artificial Intelligence

- Implement AI and ML techniques for predictive Cluster Head selection and adaptive routing decisions.
- Enhance network intelligence through data-driven optimization.

2. Hybrid Optimization Techniques

- Combine Genetic Algorithms with advanced optimization methods such as:
 - Particle Swarm Optimization (PSO)
 - Ant Colony Optimization (ACO)
 - Reinforcement Learning (RL)
- Improve routing efficiency and energy management.

3. Real-Time IoT Hardware Implementation

- Deploy and validate the protocol on practical IoT platforms, including:
 - ESP32
 - Arduino
 - LoRaWAN
 - ZigBee-based sensor networks
- Evaluate performance under real-world conditions.

4. Advanced Energy Harvesting Integration

- Incorporate multiple renewable energy sources such as:
 - Solar energy
 - Wind energy
 - Vibration energy
 - Hybrid energy harvesting systems
- Increase network sustainability and operational lifetime.

5. Support for Mobile Wireless Sensor Networks

- Extend the protocol to mobile sensor environments.
- Enable UAV-assisted data collection and communication systems.

6. Enhanced Security Mechanisms

- Integrate advanced security solutions, including:
 - Blockchain technology
 - Trust-based routing

- Intrusion Detection Systems (IDS)
- Improve data integrity, confidentiality, and network resilience.

7. Application in Diverse IoT Domains

- Deploy the protocol in various real-world applications such as:
 - Smart Agriculture
 - Smart Cities
 - Healthcare Monitoring
 - Environmental Monitoring
 - Disaster Management
 - Industrial IoT (IIoT)

8. Cloud-Based Monitoring and Management

- Develop a cloud-connected dashboard for:
 - Real-time monitoring
 - Data visualization
 - Analytics and reporting
 - Remote network management
- Facilitate efficient decision-making and system control.

Expected Benefits

- Improved network reliability and stability.
- Enhanced scalability for large-scale deployments.
- Increased intelligence through adaptive and predictive mechanisms.
- Better energy efficiency and resource utilization.
- Greater sustainability for next-generation IoT ecosystems.
- Improved security and robustness against network threats.

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Authors: Thiemo Voigt, Hartmut Ritter, Jochen Schiller

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Authors:

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Authors:

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Journal: Peer-to-Peer Networking and Applications (Springer) — **Year:** 2022 — **DOI:** 10.1007/s12083-022-01405-5

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Authors: Mohamed Hadi Habaebi, Abdullah Ahmed S. Basaloom, Md. Rafiqul Islam, Yacine Merrad, M. Mesri, Elfatih A. A. Elsheikh, F. M. Suliman

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Journal: International Journal of Distributed Sensor Networks (SAGE) — **Year:** 2021 — **DOI:** 10.1177/15501477211028549

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Authors:

Paper Title: Routing with Renewable Energy Management in Wireless Sensor Networks

Journal: Sensors (MDPI), 21(13):4376 — **Year:** 2021 — **DOI:** 10.3390/s21134376

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Authors:

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Journal: EURASIP Journal on Wireless Communications and Networking (Springer) — **Year:** 2020 — **DOI:** 10.1186/s13638-020-01721-5

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Authors:

Paper Title: Energy-Efficient Multi-Static Sink Model for Wireless Sensor Network

Journal: The Journal of Supercomputing (Springer) — **Year:** 2025 — **DOI:** 10.1007/s11227-025-07418-0

[24] (Applied Sciences, 2023)

Authors:

Paper Title: Environment-Aware Energy Efficient and Reliable Routing in Real-Time Multi-Sink Wireless Sensor Networks for Smart Cities Applications

Journal: Applied Sciences (MDPI), 13(1):605 — **Year:** 2023 — **DOI:** 10.3390/app13010605

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Authors:

Paper Title: An Energy Efficient Distance-Aware Routing Algorithm with Multiple Mobile Sinks for Wireless Sensor Networks

Journal: Sensors (MDPI), 14(8):15163 — **Year:** 2014 — **DOI:** 10.3390/s140815163

[26] (Archives of Computational Methods in Engineering, 2024)

Authors:

Paper Title: Energy-Efficient Routing in Wireless Sensor Networks: A Meta-Heuristic and Artificial-Intelligence-Based Approach—A Comprehensive Review

Journal: Archives of Computational Methods in Engineering (Springer) — **Year:** 2024 — **DOI:** 10.1007/s11831-023-10039-6

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Authors:

Paper Title: Survey of Extended LEACH-Based Clustering Routing Protocols for Wireless Sensor Networks

Source: arXiv Preprint — **Year:** 2012 — **ID:** arXiv:1207.2609

[28] (Wireless Networks, 2021)

Authors:

Paper Title: Internet of Things Multi-Hop Energy-Efficient Cluster-Based Routing Using Particle Swarm Optimization

Journal: Wireless Networks (Springer) — **Year:** 2021 — **DOI:** 10.1007/s11276-021-02801-0

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Authors: M. Pitchai, A. Shamsiev, A. Mukhamadiyev, H. Khujamatov, et al.

Paper Title: Clustered Routing Using Chaotic Genetic Algorithm with Grey Wolf Optimization to Enhance Energy Efficiency in Sensor Networks

Journal: Sensors (MDPI) — **Year:** 2024 — **DOI:** 10.3390/s24134406

Appendix (Program Code)

```
""""
=====
=====
    Solar-Aware Genetic Algorithm Based Multi-Sink Data Aggregation Protocol
    for Wireless Sensor Assisted IoT Networks
    --- GA-ONLY RUNNER ---
=====
=====
    This file is the GA-only spin-off of `solar_ga_wsn`.
    Behaviour, formulas, GA logic, MS-CH election, and snapshot output are
    IDENTICAL to the GA branch of the original combined script. LEACH has been
    removed so this script outputs only the GA-protocol results.

    Architecture (3-Tier):
    Tier 1 -> Sensor nodes : sense data, send to nearest CH
    Tier 2 -> Cluster Heads: aggregate sensors data, then either:
        PATH A -> direct to BS (close + healthy battery)
        PATH B -> via MS-CH (far or low battery)
    Tier 3 -> MS-CH node(s): collect from relay CHs, aggregate, send to BS
=====
=====
""""

import math
import os
import random
from dataclasses import dataclass, field as dc_field
from typing import List, Optional, Tuple, Dict

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.lines as mlines

# Headless-friendly: never block on a GUI, even if matplotlib is configured for one.
plt.switch_backend("Agg")

SNAPSHOT_DIR = "topology_snapshots"

#
=====
=====
# SECTION 1 - USER INPUT
#
```

```
=====
=====
```

```
def get_user_input() -> dict:
```

```
    """
```

```
    Collect simulation parameters. Press ENTER to accept the default.
```

```
    """
```

```
    print("\n" + "=" * 65)
```

```
    print(" Solar-Aware GA Multi-Sink WSN Protocol - Configuration")
```

```
    print("=" * 65)
```

```
    print(" Press ENTER to use default value shown in [brackets]\n")
```

```
def ask(prompt, default, cast=float, vmin=None, vmax=None):
```

```
    while True:
```

```
        try:
```

```
            raw = input(f" {prompt} [{default}]: ").strip()
```

```
            val = cast(raw) if raw else default
```

```
            if vmin is not None and val < vmin:
```

```
                print(f" ! Must be >= {vmin}")
```

```
                continue
```

```
            if vmax is not None and val > vmax:
```

```
                print(f" ! Must be <= {vmax}")
```

```
                continue
```

```
            return val
```

```
        except ValueError:
```

```
            print(f" ! Invalid - expected {cast.__name__}")
```

```
print(" -- Network Setup --")
```

```
field = ask("Field size in metres (square)", 100, int, 50, 2000)
```

```
n_nodes = ask("Number of sensor nodes", 50, int, 10, 5000)
```

```
bs_x = ask("Base station X position (metres)", field // 2, float, 0)
```

```
bs_y = ask("Base station Y position (metres)", field + 20, float, 0)
```

```
n_rounds = ask("Number of simulation rounds", 300, int, 50, 5000)
```

```
print("\n -- CH / MS-CH Percentages (both scale with alive nodes) --")
```

```
ch_pct = ask("CH percentage of alive nodes (e.g. 10 = 10%)",
```

```
10, float, 2, 30)
```

```
relay_per_ms = ask("Relay CHs handled by ONE MS-CH",
```

```
4, int, 1, 50)
```

```
print("\n -- Energy Settings --")
```

```
e_init = ask("Initial node energy (Joules)", 0.5, float, 0.01, 10.0)
```

```
e_solar = ask("Peak solar harvest rate (J/round)", 0.002, float, 0.0, 0.1)
```

```
pkt = ask("Packet size (bits)", 4000, int, 100, 100000)
```

```
print("\n -- GA Settings --")
```

```
ga_pop = ask("GA population size", 30, int, 10, 300)
```

```

ga_gen = ask("GA generations per round", 50, int, 10, 500)
ga_mut = ask("GA mutation rate (0.0-1.0)", 0.1, float, 0.0, 1.0)
ga_cx = ask("GA crossover rate (0.0-1.0)", 0.8, float, 0.0, 1.0)

```

```

print("\n -- Routing Thresholds --")
d_dist = ask("Max distance (m) for CH->BS direct",
             round(field * 0.75), float, 10)
d_nrg = ask("Min battery fraction for CH->BS direct (0.0-1.0)",
            round(e_init / BATTERY_MAX * 0.6, 2), float, 0.0, 1.0)

```

```

print("\n -- Topology Snapshots --")
snap_every = ask("Save topology image every N rounds (0 = off)",
                 50, int, 0, 10000)

```

```

print("\n -- Validation --")
init_chs = max(1, round(n_nodes * ch_pct / 100))
print(f" v Initial CH count : {init_chs} "
      f" (= {n_nodes} nodes x {ch_pct}%)" )
print(f" v CH and MS-CH counts auto-scale every round")
if bs_y > field * 2:
    print(" ! Warning: BS is very far - high energy cost")
print(" v Configuration accepted\n")

```

```

return {
    "FIELD" : field,
    "NUM_NODES" : n_nodes,
    "BS_X" : bs_x,
    "BS_Y" : bs_y,
    "NUM_ROUNDS" : n_rounds,
    "CH_PERCENT" : ch_pct / 100.0,
    "RELAYS_PER_MS": relay_per_ms,
    "E_INITIAL" : e_init,
    "MAX_HARVEST" : e_solar,
    "PACKET_SIZE" : pkt,
    "GA_POP" : ga_pop,
    "GA_GEN" : ga_gen,
    "GA_MUT" : ga_mut,
    "GA_CX" : ga_cx,
    "DIRECT_DIST" : d_dist,
    "DIRECT_NRG" : d_nrg,
    "SNAPSHOT_EVERY": snap_every,
}

```

```

def get_num_chs(alive_count: int, cfg: dict) -> int:
    """Dynamic CH count - recalculated every round."""
    if alive_count <= 0:

```

```

        return 0
    return max(1, min(alive_count, round(alive_count * cfg["CH_PERCENT"])))

def get_num_ms(num_relay_chs: int, cfg: dict) -> int:
    """
    Dynamic MS-CH count. One MS-CH per RELAYS_PER_MS relay CHs.
    Returns 0 when no CH needs a relay (MS-CH stage gets skipped wholesale).
    """
    if num_relay_chs <= 0:
        return 0
    return max(1, min(num_relay_chs,
        math.ceil(num_relay_chs / cfg["RELAYS_PER_MS"])))

#
=====
=====
# SECTION 2 - RADIO & ENERGY CONSTANTS
#
=====
=====

E_ELEC      = 50e-9
E_AMP       = 100e-12
E_MP        = 0.0013e-12
E_DA        = 5e-9
BATTERY_MAX = 2.0
COMM_RANGE_PCT = 0.4
D0          = math.sqrt(E_AMP / E_MP)
MS_REELECT_THR = 0.15

#
=====
=====
# SECTION 3 - NODE
#
=====
=====

class Node:
    __slots__ = ("id", "x", "y", "energy", "alive", "role",
        "assigned_ch", "assigned_ms", "goes_direct",
        "packets_sent", "_cfg", "_dist_bs")

    def __init__(self, node_id: int, x: float, y: float, cfg: dict):

```

```

self.id      = node_id
self.x       = x
self.y       = y
self.energy  = cfg["E_INITIAL"]
self.alive   = True
self.role    = "sensor"
self.assigned_ch = None
self.assigned_ms = None
self.goes_direct = False
self.packets_sent = 0
self._cfg    = cfg
self._dist_bs = math.hypot(cfg["BS_X"] - x, cfg["BS_Y"] - y)

def distance_to(self, x: float, y: float) -> float:
    return math.hypot(self.x - x, self.y - y)

@property
def distance_to_bs(self) -> float:
    return self._dist_bs

def _tx_cost(self, bits: int, d: float) -> float:
    if d <= D0:
        return E_ELEC * bits + E_AMP * bits * d * d
    d2 = d * d
    return E_ELEC * bits + E_MP * bits * d2 * d2

def _rx_cost(self, bits: int) -> float:
    return E_ELEC * bits

def _agg_cost(self, bits: int, n: int) -> float:
    return E_DA * bits * n

def transmit(self, to_x: float, to_y: float) -> bool:
    cost = self._tx_cost(self._cfg["PACKET_SIZE"],
                        math.hypot(self.x - to_x, self.y - to_y))
    self.energy -= cost
    self.packets_sent += 1
    if self.energy <= 0:
        self.energy = 0
        self.alive = False
    return self.alive

def receive(self) -> bool:
    self.energy -= self._rx_cost(self._cfg["PACKET_SIZE"])
    if self.energy <= 0:
        self.energy = 0
        self.alive = False

```

```

    return self.alive

def aggregate(self, n_packets: int) -> bool:
    self.energy -= self._agg_cost(self._cfg["PACKET_SIZE"], n_packets)
    if self.energy <= 0:
        self.energy = 0
        self.alive = False
    return self.alive

def harvest_solar(self, solar_rate_now: float) -> None:
    if solar_rate_now <= 0:
        return
    harvest = max(0.0, solar_rate_now + random.gauss(0, solar_rate_now * 0.05))
    self.energy = min(self.energy + harvest, BATTERY_MAX)

def reset_role(self) -> None:
    self.role = "sensor"
    self.assigned_ch = None
    self.assigned_ms = None
    self.goes_direct = False

@property
def energy_fraction(self) -> float:
    return min(self.energy / BATTERY_MAX, 1.0)

def __repr__(self) -> str:
    return (f"Node({self.id}, {self.role}, "
            f"{self.energy:.4f}J, alive={self.alive})")

def solar_rate_for_round(round_num: int, max_harvest: float) -> float:
    """
    24-hour cycle:
    00:00 -> 0
    06:00 -> sunrise
    12:00 -> peak solar
    18:00 -> sunset
    18:00-24:00 -> 0
    """
    hour = round_num % 24
    return max_harvest * max(
        0.0,
        math.sin(math.pi * (hour - 6) / 12)
    )

#
=====
=====

```

```
# SECTION 4 - WORLD STATE
```

```
#
```

```
=====
```

```
@dataclass
```

```
class World:
```

```
    nodes: List[Node]
```

```
    alive_idx: np.ndarray = dc_field(default_factory=lambda: np.empty(0, dtype=np.int32))
```

```
    alive_xy: np.ndarray = dc_field(default_factory=lambda: np.empty((0, 2)))
```

```
    alive_e: np.ndarray = dc_field(default_factory=lambda: np.empty(0))
```

```
    alive_dbs: np.ndarray = dc_field(default_factory=lambda: np.empty(0))
```

```
    id_to_idx: Dict[int, int] = dc_field(default_factory=dict)
```

```
def refresh(self) -> None:
```

```
    alive = [n for n in self.nodes if n.alive]
```

```
    if not alive:
```

```
        self.alive_idx = np.empty(0, dtype=np.int32)
```

```
        self.alive_xy = np.empty((0, 2))
```

```
        self.alive_e = np.empty(0)
```

```
        self.alive_dbs = np.empty(0)
```

```
        self.id_to_idx = {}
```

```
        return
```

```
    self.alive_idx = np.fromiter((n.id for n in alive), dtype=np.int32,  
                                count=len(alive))
```

```
    self.alive_xy = np.array([(n.x, n.y) for n in alive], dtype=np.float64)
```

```
    self.alive_e = np.fromiter((n.energy for n in alive),  
                               dtype=np.float64, count=len(alive))
```

```
    self.alive_dbs = np.fromiter((n._dist_bs for n in alive),  
                                 dtype=np.float64, count=len(alive))
```

```
    self.id_to_idx = {int(nid): i for i, nid in enumerate(self.alive_idx)}
```

```
def vectorized_assign(sensor_xy: np.ndarray, ch_xy: np.ndarray) -> np.ndarray:
```

```
    if ch_xy.shape[0] == 0 or sensor_xy.shape[0] == 0:
```

```
        return np.empty(sensor_xy.shape[0], dtype=np.int64)
```

```
    diff = sensor_xy[:, None, :] - ch_xy[None, :, :]
```

```
    d2 = np.einsum("ijk,ijk->ij", diff, diff)
```

```
    return d2.argmax(axis=1)
```

```
#
```

```
=====
```

```
# SECTION 5 - GENETIC ALGORITHM
```

```
#
```

```
=====
```

=====

```
class Chromosome:
    __slots__ = ("genes", "fitness")

    def __init__(self, genes):
        self.genes = list(genes)
        self.fitness = -1.0

    def copy(self) -> "Chromosome":
        c = Chromosome(self.genes)
        c.fitness = self.fitness
        return c

def _evaluate_population(pop: List["Chromosome"],
                        world: World,
                        cfg: dict,
                        solar_now: float,
                        num_chs: int) -> None:
    pending = [c for c in pop if c.fitness < 0]
    if not pending:
        return

    if (world.alive_idx.size == 0 or num_chs == 0
        or world.alive_xy.shape[0] < num_chs):
        for c in pending:
            c.fitness = 0.0
        return

    alive_xy = world.alive_xy
    alive_e = world.alive_e
    S = alive_xy.shape[0]
    id_map = world.id_to_idx
    K = num_chs
    P = len(pending)

    gene_idx = np.empty((P, K), dtype=np.int64)
    valid = np.ones(P, dtype=bool)
    for p, c in enumerate(pending):
        try:
            gene_idx[p] = [id_map[g] for g in c.genes]
        except (KeyError, ValueError):
            valid[p] = False
            gene_idx[p] = 0

    ch_xy = alive_xy[gene_idx]
```

```

ch_e = alive_e[gene_idx]

e_score = np.minimum(ch_e.sum(axis=1) / (K * cfg["E_INITIAL"]), 1.0)

if cfg["MAX_HARVEST"] > 0:
    solar_fraction = solar_now / cfg["MAX_HARVEST"]

    energy_fraction = np.minimum(
        ch_e.mean(axis=1) / cfg["E_INITIAL"],
        1.0
    )

    s_score = (
        0.5 * solar_fraction +
        0.5 * energy_fraction
    )
else:
    s_score = np.full(P, 0.5)

comm_range2 = (cfg["FIELD"] * COMM_RANGE_PCT) ** 2
big_alloc = P * S * K
if big_alloc <= 4_000_000:
    diff = alive_xy[None, :, None, :] - ch_xy[:, None, :, :]
    d2 = (diff * diff).sum(axis=-1)
    d2_min = d2.min(axis=2)
    is_ch = np.zeros((P, S), dtype=bool)
    rows = np.arange(P)[:, None]
    is_ch[rows, gene_idx] = True
    sensor_mask = ~is_ch
    within = (d2_min <= comm_range2) & sensor_mask
    denom = np.maximum(sensor_mask.sum(axis=1), 1)
    c_score = within.sum(axis=1) / denom
else:
    c_score = np.zeros(P)
    for p in range(P):
        if not valid[p]:
            continue
        mask = np.ones(S, dtype=bool)
        mask[gene_idx[p]] = False
        sxy = alive_xy[mask]
        if sxy.shape[0] == 0:
            c_score[p] = 1.0
            continue
        diff = sxy[:, None, :] - ch_xy[p][None, :, :]
        d2 = (diff * diff).sum(axis=-1)
        covered = int((d2.min(axis=1) <= comm_range2).sum())
        c_score[p] = covered / sxy.shape[0]

```

```

if K > 1:
    centroid = ch_xy.mean(axis=1, keepdims=True)
    spread = np.sqrt(((ch_xy - centroid) ** 2).sum(axis=2)).mean(axis=1)
    sp_score = np.minimum(spread / cfg["FIELD"], 1.0)
else:
    sp_score = np.full(P, 0.5)

# BS-proximity score: reward chromosome sets where at least some CHs
# are close to the BS (enabling PATH A direct routing).
bs_xy = np.array([[cfg["BS_X"], cfg["BS_Y"]]], dtype=np.float64)
max_dist = math.hypot(cfg["FIELD"], cfg["BS_Y"])
ch_bs_d = np.sqrt(((ch_xy - bs_xy[:, None, :]) ** 2).sum(axis=2)) # (P, K)
# Score = fraction of CHs that are within DIRECT_DIST of BS
bs_score = (ch_bs_d <= cfg["DIRECT_DIST"]).sum(axis=1) / max(K, 1)
bs_score = bs_score.astype(np.float64)

fitness = (0.20 * e_score + 0.20 * s_score
           + 0.30 * c_score + 0.15 * sp_score + 0.15 * bs_score)
fitness = np.where(valid, fitness, 0.0)

for p, c in enumerate(pending):
    c.fitness = float(fitness[p])

def _smart_initial_population(alive_ids: List[int],
                             alive_energies: List[float],
                             num_chs: int,
                             pop_size: int) -> List[Chromosome]:
    pop: List[Chromosome] = []
    if not alive_ids or num_chs > len(alive_ids):
        return pop
    energies = np.asarray(alive_energies, dtype=np.float64)
    weights = energies + 1e-9
    weights = weights / weights.sum()

    n_smart = pop_size // 2
    for _ in range(n_smart):
        chosen = np.random.choice(alive_ids, size=num_chs,
                                  replace=False, p=weights)
        pop.append(Chromosome(chosen.tolist()))
    while len(pop) < pop_size:
        pop.append(Chromosome(random.sample(alive_ids, num_chs)))
    return pop

def _tournament(pop: List[Chromosome], k: int = 3) -> Chromosome:

```

```

if len(pop) <= k:
    return max(pop, key=lambda c: c.fitness)
return max(random.sample(pop, k), key=lambda c: c.fitness)

def _crossover(p1: Chromosome, p2: Chromosome,
               cfg: dict, num_chs: int) -> Chromosome:
    if random.random() > cfg["GA_CX"] or num_chs < 2:
        return p1.copy()
    pt = random.randint(1, num_chs - 1)
    seen = set(p1.genes[:pt])
    genes = list(p1.genes[:pt])
    for g in p2.genes:
        if g not in seen:
            genes.append(g); seen.add(g)
            if len(genes) == num_chs:
                break
    if len(genes) < num_chs:
        for g in p1.genes:
            if g not in seen:
                genes.append(g); seen.add(g)
                if len(genes) == num_chs:
                    break
    return Chromosome(genes[:num_chs])

def _mutate(chromo: Chromosome, alive_ids: List[int],
            mut_rate: float) -> Chromosome:
    if random.random() >= mut_rate or not chromo.genes:
        return chromo
    genes = chromo.genes[:]
    gset = set(genes)
    candidates = [i for i in alive_ids if i not in gset]
    if not candidates:
        return Chromosome(genes)
    idx = random.randint(0, len(genes) - 1)
    genes[idx] = random.choice(candidates)
    return Chromosome(genes)

def run_ga_ch_election(nodes: List[Node],
                       world: World,
                       cfg: dict,
                       round_num: int,
                       num_chs: int) -> Optional[Chromosome]:
    if world.alive_idx.size < num_chs:
        return None

```

```

alive_ids = world.alive_idx.tolist()
alive_e = world.alive_e.tolist()
solar_now = solar_rate_for_round(round_num, cfg["MAX_HARVEST"])

pop = _smart_initial_population(alive_ids, alive_e, num_chs, cfg["GA_POP"])
if not pop:
    return None

PATIENCE = max(5, cfg["GA_GEN"] // 8)
best_overall: Optional[Chromosome] = None
best_score = -1.0
stale = 0
cur_mut = cfg["GA_MUT"]

for gen in range(cfg["GA_GEN"]):
    _evaluate_population(pop, world, cfg, solar_now, num_chs)
    pop.sort(key=lambda c: c.fitness, reverse=True)

    if pop[0].fitness > best_score + 1e-9:
        best_score = pop[0].fitness
        best_overall = pop[0].copy()
        stale = 0
        cur_mut = cfg["GA_MUT"]
    else:
        stale += 1
        cur_mut = min(0.5, cfg["GA_MUT"] * (1 + stale * 0.25))

    if stale >= PATIENCE:
        break

    new_pop = [pop[0].copy()]
    if len(pop) > 1:
        new_pop.append(pop[1].copy())

    while len(new_pop) < cfg["GA_POP"]:
        child = _crossover(_tournament(pop), _tournament(pop),
                           cfg, num_chs)
        new_pop.append(_mutate(child, alive_ids, cur_mut))
    pop = new_pop

return best_overall

```

#

```

=====
=====

```

SECTION 6 - PATH DECISION

#

=====

```
def decide_ch_paths(ch_nodes: List[Node],
                    cfg: dict) -> Tuple[List[Node], List[Node]]:
    direct_chs: List[Node] = []
    relay_chs: List[Node] = []
    for ch in ch_nodes:
        if not ch.alive:
            continue
        close = ch.distance_to_bs <= cfg["DIRECT_DIST"]
        healthy = ch.energy_fraction >= cfg["DIRECT_NRG"]
        if close and healthy:
            ch.goes_direct = True
            ch.assigned_ms = None
            direct_chs.append(ch)
        else:
            ch.goes_direct = False
            ch.assigned_ms = None
            relay_chs.append(ch)
    return direct_chs, relay_chs
```

#

=====

SECTION 7 - MS-CH ELECTION (THE SOLAR-AWARE NOVELTY)

#

=====

```
def _solar_aware_score(ch: Node, peers: List[Node], cfg: dict,
                      solar_now: float) -> float:
    bat = min(ch.energy_fraction, 1.0)
    if cfg["MAX_HARVEST"] > 0:
        solar = solar_now / cfg["MAX_HARVEST"]
    else:
        solar = 0.5

    if len(peers) > 1:
        d_sum = sum(ch.distance_to(p.x, p.y) for p in peers if p.id != ch.id)
        avg_d = d_sum / max(len(peers) - 1, 1)
        cent = 1.0 - min(avg_d / (cfg["FIELD"] * math.sqrt(2)), 1.0)
    else:
        cent = 1.0
```

```

max_d = math.sqrt(cfg["FIELD"] ** 2 + cfg["BS_Y"] ** 2)
bs_s = 1.0 - min(ch.distance_to_bs / max_d, 1.0)

return 0.35 * bat + 0.30 * solar + 0.20 * cent + 0.15 * bs_s

```

```

def _kmedoids_split(relay_chs: List[Node], k: int,
                  iters: int = 10) -> List[List[Node]]:
    if k <= 1 or len(relay_chs) <= k:
        if k <= 1:
            return [relay_chs]
        return [[ch] for ch in relay_chs]

    xy = np.array([(c.x, c.y) for c in relay_chs], dtype=np.float64)
    n = len(relay_chs)
    medoids = [random.randrange(n)]
    for _ in range(k - 1):
        d2 = np.min(((xy[:, None, :] - xy[medoids][None, :, :]) ** 2)
                    .sum(axis=2), axis=1)
        medoids.append(int(d2.argmax()))

    labels = np.zeros(n, dtype=np.int32)
    for _ in range(iters):
        diff = xy[:, None, :] - xy[medoids][None, :, :]
        d2 = np.einsum("ijk,ijk->ij", diff, diff)
        new_labels = d2.argmax(axis=1)
        if np.array_equal(new_labels, labels) and _ > 0:
            break
        labels = new_labels
        for ki in range(k):
            members = np.where(labels == ki)[0]
            if members.size == 0:
                continue
            centroid = xy[members].mean(axis=0)
            d_to_c = ((xy[members] - centroid) ** 2).sum(axis=1)
            medoids[ki] = int(members[int(d_to_c.argmin())])

    clusters: List[List[Node]] = [[] for _ in range(k)]
    for i, lab in enumerate(labels):
        clusters[lab].append(relay_chs[i])
    return [c for c in clusters if c]

```

```

def elect_ms_chs(relay_chs: List[Node], cfg: dict,
                 solar_now: float, num_ms: int) -> List[Node]:
    if num_ms <= 0 or not relay_chs:

```

```

    return []

if num_ms == 1 or len(relay_chs) <= 2:
    best = max(relay_chs,
               key=lambda c: _solar_aware_score(c, relay_chs, cfg, solar_now))
    best.role = "MS-CH"
    for r in relay_chs:
        if r.id != best.id:
            r.assigned_ms = best.id
    return [best]

clusters = _kmedoids_split(relay_chs, num_ms)
elected: List[Node] = []
for cluster in clusters:
    best = max(cluster,
               key=lambda c: _solar_aware_score(c, cluster, cfg, solar_now))
    best.role = "MS-CH"
    elected.append(best)
    for r in cluster:
        if r.id != best.id:
            r.assigned_ms = best.id

return elected

#
=====
=====
# SECTION 8 - GA PROTOCOL ROUND
#
=====
=====

def simulate_round_ga(nodes: List[Node],
                     world: World,
                     round_num: int,
                     cfg: dict,
                     stats: dict) -> bool:
    solar_now = solar_rate_for_round(round_num, cfg["MAX_HARVEST"])

    if solar_now > 0:
        for n in nodes:
            if n.alive:
                n.harvest_solar(solar_now)

    for n in nodes:
        if n.alive:

```

```

        n.reset_role()

world.refresh()
alive_count = world.alive_idx.size
num_chs = get_num_chs(alive_count, cfg)

if alive_count < num_chs + 1:
    _record_stats(nodes, stats)
    stats["ch_counts"].append(0)
    stats["ms_counts"].append(0)
    return False

solution = run_ga_ch_election(nodes, world, cfg, round_num, num_chs)
if solution is None:
    _record_stats(nodes, stats)
    stats["ch_counts"].append(0)
    stats["ms_counts"].append(0)
    return False

ch_nodes = [nodes[g] for g in solution.genes
             if g < len(nodes) and nodes[g].alive]
for ch in ch_nodes:
    ch.role = "CH"

direct_chs, relay_chs = decide_ch_paths(ch_nodes, cfg)

num_ms = get_num_ms(len(relay_chs), cfg)
ms_chs = elect_ms_chs(relay_chs, cfg, solar_now, num_ms) if num_ms else []

if ch_nodes:
    ch_xy = np.array([(c.x, c.y) for c in ch_nodes], dtype=np.float64)
    ch_id_arr = np.array([c.id for c in ch_nodes], dtype=np.int64)
    ch_id_set = {c.id for c in ch_nodes}
    sensor_mask = np.array([nid not in ch_id_set
                           for nid in world.alive_idx], dtype=bool)
    sensor_xy = world.alive_xy[sensor_mask]
    sensor_ids = world.alive_idx[sensor_mask]
    if sensor_xy.shape[0] > 0:
        assign_idx = vectorized_assign(sensor_xy, ch_xy)
        for sid, ai in zip(sensor_ids.tolist(), assign_idx.tolist()):
            nodes[sid].assigned_ch = int(ch_id_arr[ai])

for nid in world.alive_idx.tolist():
    n = nodes[nid]
    if n.role == "sensor" and n.assigned_ch is not None:
        ch = nodes[n.assigned_ch]
        if ch.alive:

```

```

        n.transmit(ch.x, ch.y)
        ch.receive()

member_count: Dict[int, int] = {c.id: 0 for c in ch_nodes}
for nid in world.alive_idx.tolist():
    n = nodes[nid]
    if n.role == "sensor" and n.assigned_ch in member_count:
        member_count[n.assigned_ch] += 1

ms_inbound: Dict[int, int] = {m.id: 0 for m in ms_chs}

for ch in ch_nodes:
    if not ch.alive or ch.role != "CH":
        continue
    ch.aggregate(member_count[ch.id])
    if not ch.alive:
        continue
    if ch.goes_direct:
        ch.transmit(cfg["BS_X"], cfg["BS_Y"])
    elif ch.assigned_ms is not None and ch.assigned_ms in ms_inbound:
        ms = nodes[ch.assigned_ms]
        if ms.alive:
            ch.transmit(ms.x, ms.y)
            ms.receive()
            ms_inbound[ms.id] += 1

for ms in list(ms_chs):
    if ms.energy < MS_REELECT_THR * cfg["E_INITIAL"]:
        cluster = [c for c in relay_chs
                    if c.alive and c.id != ms.id and c.assigned_ms == ms.id]
        if cluster:
            ms.role = "CH"
            new_ms = max(cluster, key=lambda c:
                          _solar_aware_score(c, cluster, cfg, solar_now))
            new_ms.role = "MS-CH"
            for r in cluster:
                if r.id != new_ms.id:
                    r.assigned_ms = new_ms.id
            ms_inbound[new_ms.id] = ms_inbound.pop(ms.id, 0)
            ms_chs[ms_chs.index(ms)] = new_ms
            stats["reelections"] += 1

for ms in ms_chs:
    if ms.alive and ms.role == "MS-CH":
        own_members = member_count.get(ms.id, 0)
        inbound = ms_inbound.get(ms.id, 0)
        ms.aggregate(own_members + inbound)

```

```

        if ms.alive:
            ms.transmit(cfg["BS_X"], cfg["BS_Y"])
            stats["packets_to_bs"] += 1

for ch in direct_chs:
    if ch.alive:
        stats["packets_to_bs"] += 1

_record_stats(nodes, stats)
stats["ch_counts"].append(len(ch_nodes))
stats["ms_counts"].append(len(ms_chs))

snap = cfg.get("SNAPSHOT_EVERY", 0)
if snap and round_num % snap == 0:
    try:
        plot_topology_snapshot(nodes, ch_nodes, direct_chs, relay_chs,
                               ms_chs, round_num, cfg, protocol="GA")
    except Exception as e:
        print(f" ! Snapshot failed at round {round_num}: {e}")
    return any(n.alive for n in nodes)

#
=====
=====
# SECTION 9 - STATS
#
=====
=====

def make_stats() -> dict:
    return {
        "alive_nodes" : [],
        "dead_nodes" : [],
        "total_energy" : [],
        "energy_stddev" : [],
        "ch_counts" : [],
        "ms_counts" : [],
        "packets_to_bs" : 0,
        "reelections" : 0,
    }

def _record_stats(nodes: List[Node], stats: dict) -> None:
    energies = [n.energy for n in nodes if n.alive]
    alive = len(energies)
    stats["alive_nodes"].append(alive)

```

```

stats["dead_nodes"].append(len(nodes) - alive)
stats["total_energy"].append(sum(energies))
stats["energy_stddev"].append(float(np.std(energies)) if energies else 0.0)

#
=====
=====
# SECTION 10 - SIMULATION RUNNER
#
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=====
def create_nodes(cfg: dict, seed=None) -> List[Node]:
    if seed is not None:
        random.seed(seed)
        np.random.seed(seed)

    return [Node(
        i,
        random.uniform(0, cfg["FIELD"]),
        random.uniform(0, cfg["FIELD"]),
        cfg
    ) for i in range(cfg["NUM_NODES"])]

def run_simulation(cfg: dict):
    nodes = create_nodes(cfg, seed=42)
    world = World(nodes=nodes)
    stats = make_stats()
    init_chs = max(1, round(cfg["NUM_NODES"] * cfg["CH_PERCENT"]))
    print(f"\n{'=' * 58}")
    print(f" Protocol : GA (Solar-Aware Multi-Sink)")
    print(f" Nodes    : {cfg['NUM_NODES']} "
          f"Field : {cfg['FIELD']}x{cfg['FIELD']}m")
    print(f" Rounds   : {cfg['NUM_ROUNDS']} "
          f"Initial CHs: {init_chs} "
          f"({cfg['CH_PERCENT'] * 100:.0f}% - dynamic)")
    print(f" MS-CH    : 1 per {cfg['RELAYS_PER_MS']} relay CHs (dynamic, "
          f"0 if no relays)")
    if cfg.get("SNAPSHOT_EVERY", 0):
        print(f" Snapshots : every {cfg['SNAPSHOT_EVERY']} rounds -> "
              f"./{SNAPSHOT_DIR}")
    print(f"{'=' * 58}")

    first_dead = None
    network_dead = cfg["NUM_ROUNDS"]

```

```

for r in range(cfg["NUM_ROUNDS"]):
    alive = simulate_round_ga(nodes, world, r, cfg, stats)

    dead_now = cfg["NUM_NODES"] - sum(1 for n in nodes if n.alive)
    if first_dead is None and dead_now >= 1:
        first_dead = r
        print(f" * First node died   : Round {r}")

    if not alive:
        network_dead = r
        print(f" X Network dead     : Round {r}")
        break

    step = max(cfg["NUM_ROUNDS"] // 5, 1)
    if r % step == 0:
        a = stats["alive_nodes"][-1]
        e = stats["total_energy"][-1]
        nchs = stats["ch_counts"][-1] if stats["ch_counts"] else "?"
        nms = stats["ms_counts"][-1] if stats["ms_counts"] else "?"
        print(f" Round {r:4d} | Alive: {a:3d}/{cfg["NUM_NODES"]} "
              f"| CHs: {nchs} | MS: {nms} | Energy: {e:.4f} J")

    if network_dead == cfg["NUM_ROUNDS"]:
        print(f" v Network survived all {cfg["NUM_ROUNDS"]} rounds!")
    print(f" Packets to BS   : {stats['packets_to_bs']}")
    print(f" MS-CH re-elections: {stats['reelections']}")
    print(f"{'=' * 58}")

    return stats, first_dead, network_dead, nodes

```

```

#
=====
=====
# SECTION 11 - PLOTS
#
=====
=====

```

```

def plot_results(ga_stats, ga_fd, cfg) -> None:
    """6 graphs: lifetime, energy, deaths, balance, dynamic CH/MS, solar."""
    fig, axes = plt.subplots(2, 3, figsize=(17, 10))
    fig.suptitle(
        "Solar-Aware GA Multi-Sink Protocol\n"
        f"(Nodes={cfg["NUM_NODES"]}, Field={cfg["FIELD"]})m, "
        f"Rounds={cfg["NUM_ROUNDS"]}, "
        f"CH%={cfg["CH_PERCENT"] * 100:.0f}% dynamic, "
    )

```

```

f"1 MS-CH per {cfg['RELAYS_PER_MS']} relay CHs)",
    fontsize=12, fontweight="bold")

C_GA, C_MS = "#1A5FAD", "#27AE60"
rg = range(len(ga_stats["alive_nodes"]))

# 1. Alive
ax = axes[0, 0]
ax.plot(rg, ga_stats["alive_nodes"], color=C_GA, lw=2, label="GA")

if ga_fd is not None:
    ax.axvline(ga_fd, color=C_GA, ls=":", alpha=0.6,
               label=f"GA 1st death r{ga_fd}")
ax.set(xlabel="Round", ylabel="Alive nodes",
       title="Network Lifetime",
       ylim=(0, cfg["NUM_NODES"] + 2))
ax.legend(fontsize=8); ax.grid(True, alpha=0.3)

# 2. Total residual energy
ax = axes[0, 1]
ax.plot(rg, ga_stats["total_energy"], color=C_GA, lw=2, label="GA")
ax.set(xlabel="Round", ylabel="Total residual energy (J)",
       title="Total Residual Energy")
ax.legend(fontsize=8); ax.grid(True, alpha=0.3)

# 3. Dead nodes
ax = axes[0, 2]
ax.plot(rg, ga_stats["dead_nodes"], color=C_GA, lw=2, label="GA")
ax.set(xlabel="Round", ylabel="Cumulative dead nodes",
       title="Node Deaths Over Time")
ax.legend(fontsize=8); ax.grid(True, alpha=0.3)

# 4. Energy balance
ax = axes[1, 0]
ax.plot(rg, ga_stats["energy_stddev"], color=C_GA, lw=2, label="GA")
ax.set(xlabel="Round", ylabel="Std deviation of energy (J)",
       title="Energy Balance\n(lower = more balanced)")
ax.legend(fontsize=8); ax.grid(True, alpha=0.3)

# 5. Dynamic CH + MS counts
ax = axes[1, 1]
if ga_stats["ch_counts"]:
    ax.plot(range(len(ga_stats["ch_counts"])),
            ga_stats["ch_counts"], color=C_GA, lw=2, label="GA CHs")
if ga_stats["ms_counts"]:
    ax.plot(range(len(ga_stats["ms_counts"])),
            ga_stats["ms_counts"], color=C_MS, lw=2, label="GA MS-CHs")

```

```

ax.set(xlabel="Round", ylabel="Count",
       title=f"Dynamic CH & MS-CH Counts\n"
             f"(both scale with alive nodes)")
ax.legend(fontsize=8); ax.grid(True, alpha=0.3)

```

6. Solar harvest cycle

```

ax = axes[1, 2]
hours = np.linspace(0, 48, 500)
harvest = [cfg["MAX_HARVEST"] * max(0, math.sin(math.pi * (h % 24) / 12))
           for h in hours]
ax.fill_between(hours, harvest, alpha=0.25, color="orange")
ax.plot(hours, harvest, color="darkorange", lw=2)
ax.set_xticks(range(0, 49, 6))
ax.set_xticklabels([f"{h % 24:02d}:00" for h in range(0, 49, 6)])
ax.set(xlabel="Hour of day", ylabel="Harvest rate (J/round)",
       title="Solar Harvest Model\n(30% weight in MS-CH election)")
ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig("ga_results.png", dpi=150, bbox_inches="tight")
print(" Plot saved -> ga_results.png")
plt.close()

```

```

def plot_topology_snapshot(nodes: List["Node"],
                          ch_nodes: List["Node"],
                          direct_chs: List["Node"],
                          relay_chs: List["Node"],
                          ms_chs: List["Node"],
                          round_num: int,
                          cfg: dict,
                          protocol: str = "GA") -> None:
    F, BSX, BSY = cfg["FIELD"], cfg["BS_X"], cfg["BS_Y"]
    os.makedirs(SNAPSHOT_DIR, exist_ok=True)

    alive_sensors_x: List[float] = []
    alive_sensors_y: List[float] = []
    dead_x: List[float] = []
    dead_y: List[float] = []
    sensor_to_ch: Dict[int, int] = {}
    ms_id_set = {m.id for m in ms_chs}
    for n in nodes:
        if not n.alive:
            dead_x.append(n.x)
            dead_y.append(n.y)
            continue
        if n.role == "sensor":

```

```

    alive_sensors_x.append(n.x)
    alive_sensors_y.append(n.y)
    if n.assigned_ch is not None:
        sensor_to_ch[n.id] = n.assigned_ch

n_alive = sum(1 for n in nodes if n.alive)
n_direct = len(direct_chs)
n_relay = len(relay_chs) - len(ms_chs)
n_ms = len(ms_chs)
ms_used = n_ms > 0 and any(c.assigned_ms is not None for c in relay_chs)

fig, ax = plt.subplots(figsize=(11, 11))
ax.set_facecolor("#f4f6f9")
ax.set_xlim(-8, F + 8)
ax.set_ylim(-8, max(BSY + 20, F + 20))

comm_range = F * COMM_RANGE_PCT
for c in ch_nodes:
    ax.add_patch(plt.Circle((c.x, c.y), comm_range,
                           color="gray", fill=False,
                           alpha=0.08, lw=0.7, ls="--"))

for sid, cid in sensor_to_ch.items():
    s, c = nodes[sid], nodes[cid]
    if c.alive:
        ax.plot([s.x, c.x], [s.y, c.y],
                color="#AABCD4", lw=0.45, alpha=0.40, zorder=1)

for c in direct_chs:
    if c.alive:
        ax.annotate("", xy=(BSX, BSY), xytext=(c.x, c.y),
                    arrowprops=dict(arrowstyle="->", color="#1A5FAD",
                                    lw=1.6, linestyle="dashed",
                                    alpha=0.85),
                    zorder=4)

for c in relay_chs:
    if c.alive and c.assigned_ms is not None and c.id not in ms_id_set:
        ms = nodes[c.assigned_ms]
        if ms.alive:
            ax.annotate("", xy=(ms.x, ms.y), xytext=(c.x, c.y),
                        arrowprops=dict(arrowstyle="->", color="#27AE60",
                                        lw=1.6, alpha=0.9),
                        zorder=4)

for m in ms_chs:
    if m.alive:

```

```

        ax.annotate("", xy=(BSX, BSY), xytext=(m.x, m.y),
                    arrowprops=dict(arrowstyle="->", color="#C0392B",
                                    lw=2.4, alpha=0.95),
                    zorder=5)

if alive_sensors_x:
    ax.scatter(alive_sensors_x, alive_sensors_y,
               c="#7FB3D3", s=32, zorder=3, alpha=0.85,
               edgecolors="white", lw=0.5)
if dead_x:
    ax.scatter(dead_x, dead_y, c="#888", s=22, marker="x",
               zorder=3, alpha=0.6, lw=1.0)

for c in direct_chs:
    if c.alive:
        ax.scatter(c.x, c.y, c="#1A5FAD", s=170, marker="^",
                   zorder=5, edgecolors="white", lw=1.2)
for c in relay_chs:
    if c.alive and c.id not in ms_id_set:
        ax.scatter(c.x, c.y, c="#E67E22", s=170, marker="^",
                   zorder=5, edgecolors="white", lw=1.2)
for m in ms_chs:
    if m.alive:
        ax.scatter(m.x, m.y, c="#27AE60", s=360, marker="*",
                   zorder=6, edgecolors="white", lw=1.4)

ax.scatter(BSX, BSY, c="#C0392B", s=420, marker="s",
           zorder=7, edgecolors="darkred", lw=2)
ax.text(BSX + 7, BSY + 3, "BS",
        fontsize=10, fontweight="bold", color="darkred")

ax.add_patch(plt.Rectangle((0, 0), F, F, fill=False,
                           edgecolor="#999", lw=1.2, ls="--", alpha=0.5))

if ms_used:
    mode_label = f"MS-USED ({n_ms} MS-CH{'s' if n_ms != 1 else ''})"
    title_color = "#27AE60"
else:
    mode_label = "MS-SKIPPED (all CHs reach BS directly)"
    title_color = "#1A5FAD"

ax.set_title(
    f"GA Round {round_num:4d} | {mode_label}\n"
    f"Alive: {n_alive}/{cfg['NUM_NODES']} "
    f"CHs: {len(ch_nodes)} "
    f"Direct: {n_direct} Relay: {n_relay} MS-CH: {n_ms}",
    fontsize=12, fontweight="bold", color=title_color, pad=12)

```

```

handles = [
    mpatches.Patch(color="#7FB3D3", label="Sensor (alive)"),
    mpatches.Patch(color="#888", label="Sensor (dead)"),
    mpatches.Patch(color="#1A5FAD", label="CH -> BS direct (PATH A)"),
    mpatches.Patch(color="#E67E22", label="CH -> MS-CH relay (PATH B)"),
    mpatches.Patch(color="#27AE60", label="MS-CH -> BS (PATH C)"),
    mpatches.Patch(color="#C0392B", label="Base Station"),
]
ax.legend(handles=handles, loc="lower right",
          fontsize=8, framealpha=0.93, edgecolor="gray")
ax.set_xlabel("X (m)")
ax.set_ylabel("Y (m)")
ax.grid(True, alpha=0.18)

fname = os.path.join(SNAPSHOT_DIR,
                     f"ga_round_{round_num:04d}.png")
plt.tight_layout()
plt.savefig(fname, dpi=120, bbox_inches="tight")
plt.close(fig)

def plot_topology_both_paths(cfg: dict) -> None:
    """Deterministic illustrative topology with multiple MS-CHs."""
    F, BSX, BSY = cfg["FIELD"], cfg["BS_X"], cfg["BS_Y"]

    def d(x1, y1, x2, y2):
        return math.hypot(x1 - x2, y1 - y2)

    sensors = [
        (0, F * .80, F * .85), (1, F * .90, F * .70), (2, F * .75, F * .72),
        (3, F * .15, F * .88), (4, F * .25, F * .75), (5, F * .10, F * .70),
        (6, F * .10, F * .35), (7, F * .20, F * .22), (8, F * .08, F * .15),
        (9, F * .80, F * .30), (10, F * .90, F * .18), (11, F * .70, F * .12),
        (12, F * .40, F * .45), (13, F * .60, F * .42), (14, F * .35, F * .30),
        (15, F * .65, F * .28),
    ]

    chs = {
        20: (F * .85, F * .92, "CH-direct", "CH-1\n(direct)"),
        21: (F * .45, F * .88, "CH-direct", "CH-2\n(direct)"),
        22: (F * .15, F * .20, "CH-relay", "CH-3\n(relay W)"),
        23: (F * .82, F * .15, "CH-relay", "CH-4\n(relay E)"),
        24: (F * .25, F * .50, "MS-CH", "MS-CH 1\n(west zone)"),
        25: (F * .75, F * .55, "MS-CH", "MS-CH 2\n(east zone)"),
    }

```

```

sensor_ch = {0: 20, 1: 20, 2: 20, 3: 21, 4: 21, 5: 21,
             6: 22, 7: 22, 8: 22, 9: 23, 10: 23, 11: 23,
             12: 24, 13: 24, 14: 25, 15: 25}
relay_to_ms = {22: 24, 23: 25}

fig, ax = plt.subplots(figsize=(11, 12))
ax.set_facecolor("#f4f6f9")
ax.set_xlim(-8, F + 8); ax.set_ylim(-8, BSY + 20)

for cid, (cx, cy, role, _) in chs.items():
    ax.add_patch(plt.Circle((cx, cy), F * COMM_RANGE_PCT,
                           color="gray", fill=False,
                           alpha=0.10, lw=0.8, ls="--"))

for sid, (_, sx, sy) in enumerate(sensors):
    chx, chy = chs[sensor_ch[sid]][1:2]
    ax.plot([sx, chx], [sy, chy],
            color="#AABCD4", lw=0.6, alpha=0.45, zorder=1)

for cid, rad, lbl in [(20, 0.20, "CH-1"), (21, -0.15, "CH-2")]:
    cx, cy = chs[cid][1:2]
    ax.annotate("", xy=(BSX, BSY), xytext=(cx, cy),
                arrowprops=dict(arrowstyle="->", color="#1A5FAD",
                                lw=2.2, linestyle="dashed",
                                connectionstyle=f"arc3,rad={rad}"),
                zorder=4)
    ax.text((cx + BSX) / 2 + (8 if rad > 0 else -8), (cy + BSY) / 2,
            f"PATH A\n{lbl}->BS\n({d(cx, cy, BSX, BSY):.0f}m)",
            fontsize=7.5, color="#1A5FAD", ha="center",
            bbox=dict(boxstyle="round,pad=0.25", fc="white",
                    ec="#1A5FAD", alpha=0.80, lw=0.7))

for r_id, ms_id in relay_to_ms.items():
    cx, cy = chs[r_id][1:2]
    mx, my = chs[ms_id][1:2]
    ax.annotate("", xy=(mx, my), xytext=(cx, cy),
                arrowprops=dict(arrowstyle="->", color="#27AE60",
                                lw=2.2,
                                connectionstyle="arc3,rad=0.15"),
                zorder=4)
    ax.text((cx + mx) / 2 + 6, (cy + my) / 2 + 4,
            f"PATH B\nCH->MS-CH\n({d(cx, cy, mx, my):.0f}m)",
            fontsize=7.5, color="#27AE60", ha="center",
            bbox=dict(boxstyle="round,pad=0.25", fc="white",
                    ec="#27AE60", alpha=0.80, lw=0.7))

for ms_id in (24, 25):

```

```

mx, my = chs[ms_id][:2]
ax.annotate("", xy=(BSX, BSY), xytext=(mx, my),
            arrowprops=dict(arrowstyle="->", color="#C0392B",
                            lw=3.0,
                            connectionstyle="arc3,rad=0.0"),
            zorder=5)
ax.text((mx + BSX) / 2 + 10, (my + BSY) / 2,
        f"PATH C\nMS->BS\n({d(mx, my, BSX, BSY):.0f}m)",
        fontsize=8, color="#C0392B", fontweight="bold", ha="center",
        bbox=dict(boxstyle="round,pad=0.3", fc="white",
                  ec="#C0392B", alpha=0.88, lw=0.9))

sx = [s[1] for s in sensors]; sy = [s[2] for s in sensors]
ax.scatter(sx, sy, c="#7FB3D3", s=45, zorder=3, alpha=0.85,
           edgecolors="white", lw=0.6)

for cid, (cx, cy, role, lbl) in chs.items():
    if role == "CH-direct":
        ax.scatter(cx, cy, c="#1A5FAD", s=210, marker="^", zorder=5,
                  edgecolors="white", lw=1.3)
    elif role == "CH-relay":
        ax.scatter(cx, cy, c="#E67E22", s=210, marker="^", zorder=5,
                  edgecolors="white", lw=1.3)
    else:
        ax.scatter(cx, cy, c="#27AE60", s=420, marker="*", zorder=6,
                  edgecolors="white", lw=1.5)
    ax.annotate(lbl, (cx, cy), textcoords="offset points",
               xytext=(9, 7), fontsize=7.5,
               bbox=dict(boxstyle="round,pad=0.2", fc="white",
                         alpha=0.75, ec="gray", lw=0.5))

ax.scatter(BSX, BSY, c="#C0392B", s=520, marker="s", zorder=7,
           edgecolors="darkred", lw=2)
ax.text(BSX + 9, BSY + 3, "Base\nStation",
        fontsize=10, fontweight="bold", color="darkred")

ax.text(0.01, 0.99,
        "Data Flow Summary\n"
        "-----\n"
        "Sensor -> CH (always)\n"
        "PATH A: CH -> BS direct\n"
        " (close + healthy batt)\n"
        "PATH B: CH -> MS-CH\n"
        " (far or low battery)\n"
        "PATH C: MS-CH -> BS\n"
        " (one long-range TX)\n"
        "Multiple MS-CHs scale\n")

```

```

        "with relay CH count.",
        transform=ax.transAxes, fontsize=8.5, va="top",
        bbox=dict(boxstyle="round,pad=0.5", fc="white",
            ec="#555", alpha=0.90, lw=0.8))

ax.add_patch(plt.Rectangle((0, 0), F, F, fill=False,
            edgecolor="#999", lw=1.5,
            ls="--", alpha=0.5))
ax.text(3, F - 5, f"{F}x{F}m sensor field",
        fontsize=8.5, color="#888")

handles = [
    mpatches.Patch(color="#7FB3D3", label="Sensor node"),
    mpatches.Patch(color="#1A5FAD", label="CH -> BS direct (close + healthy)"),
    mpatches.Patch(color="#E67E22", label="CH -> MS-CH relay (far or low batt)"),
    mpatches.Patch(color="#27AE60", label="MS-CH (solar-aware, multiple zones)"),
    mpatches.Patch(color="#C0392B", label="Base Station (BS)"),
    mlines.Line2D([], [], color="#AABCD4", lw=1.2, label="Sensor -> CH"),
    mlines.Line2D([], [], color="#1A5FAD", lw=2.2, ls="--",
        label="PATH A: CH -> BS directly"),
    mlines.Line2D([], [], color="#27AE60", lw=2.2,
        label="PATH B: CH -> MS-CH (relay)"),
    mlines.Line2D([], [], color="#C0392B", lw=3.0,
        label="PATH C: MS-CH -> BS (final hop)",
]
ax.legend(handles=handles, loc="lower right",
        fontsize=8.2, framealpha=0.93, edgecolor="gray")
ax.set_xlabel("X position (metres)", fontsize=11)
ax.set_ylabel("Y position (metres)", fontsize=11)
ax.set_title(
    "WSN Network Topology - Solar-Aware GA Multi-Sink Protocol\n"
    "PATH A: CH->BS direct | PATH B: CH->MS-CH->BS | "
    "PATH C: MS-CH->BS",
    fontsize=12, fontweight="bold")
ax.grid(True, alpha=0.18)
plt.tight_layout()
plt.savefig("ga_topology.png", dpi=160, bbox_inches="tight")
print(" Plot saved -> ga_topology.png")
plt.close()

```

#

=====

=====

SECTION 12 - SUMMARY

#

=====

```
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```

```
def print_summary(ga_s, ga_fd, ga_nd, cfg) -> None:
```

```
    print("\n" + "=" * 64)
    print(" GA SIMULATION RESULTS SUMMARY")
    print("=" * 64)
    print(f" {'Metric':<40} {'GA':>10}")
    print(f" {'-' * 50}")
```

```
def fmt(v):
    return str(v) if v is not None else f">{cfg['NUM_ROUNDS']}"
```

```
rows = [
    ("First node death (round)",    fmt(ga_fd)),
    ("Network lifetime (rounds)",   str(ga_nd)),
    ("Packets delivered to BS",     str(ga_s["packets_to_bs"])),
    ("Final residual energy (J)",
     f"{ga_s['total_energy'][-1]:.4f}" if ga_s["total_energy"] else "0"),
    ("Final energy std dev (J)",
     f"{ga_s['energy_stddev'][-1]:.4f}" if ga_s["energy_stddev"] else "0"),
    ("MS-CH re-elections",         str(ga_s["reelections"])),
    ("Avg CHs / round",
     f"{np.mean(ga_s['ch_counts']):.1f}" if ga_s['ch_counts'] else "0"),
    ("Avg MS-CHs / round",
     f"{np.mean(ga_s['ms_counts']):.2f}" if ga_s['ms_counts'] else "0"),
    ("CH% used", f"{cfg['CH_PERCENT'] * 100:.0f}% dynamic"),
]
for label, ga_val in rows:
    print(f" {label:<40} {ga_val:>10}")
print("=" * 64)
```

```
#
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```
# MAIN
```

```
#
```

```
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```

```
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```

```
def main():
```

```
    print("\n" + "#" * 65)
    print(" Solar-Aware GA Multi-Sink Data Aggregation Protocol")
    print(" for Wireless Sensor Assisted IoT -- GA-ONLY RUNNER")
    print("#" * 65)
```

```
    cfg = get_user_input()
```

```
ga_stats, ga_fd, ga_nd, _ = run_simulation(cfg)

print_summary(ga_stats, ga_fd, ga_nd, cfg)

print("\n Generating plots...")
plot_results(ga_stats, ga_fd, cfg)
plot_topology_both_paths(cfg)

print("\n All outputs saved. Done.\n")

if __name__ == "__main__":
    main()
```